

Emergence and variability of static-pattern decoded motion in a connectome-constrained visual network

Abstract

Static motion illusion-like discrete design (MIDD) patterns are physically stationary, yet can evoke directional motion readouts. We examined these responses in connectome-constrained FlyVis networks trained for optic-flow estimation without neural-response supervision. The antisymmetric component was near zero at initialization ($|A| < 3 \times 10^{-9}$) and emerged during ordinary optic-flow learning, although MIDD was not a training target. For 3MIDD C=48 across phases in the official 50-model cohort, absolute-agreement ICC(A,1) was 0.7647, and 34/50 models changed the sign of A . For a centered eight-field rendering, measured local three-level responses conditionally predicted held-out four-level scalar outputs (mean $R^2 = 0.9024$); this affine prediction used measured local responses at held-out luminances and does not imply vector-field additivity. Sector shuffles and other static controls produced comparable or larger responses; canonical 72/96-sector patterns were not normalized-response maxima in any of ten models, and the decoded output was not selective for MIDD under these tested conditions. Three model-level associations with Sintel endpoint error did not survive Holm correction. Decoder-input replacements altered both MIDD and physical-motion readouts. Network-decoder cross-swaps changed decoded magnitude and direction readout, but only 9/90 mismatched pairs correctly decoded both physical directions. Decoded flow is a model output, not perception.

Introduction

Stationary luminance patterns can elicit motion-related behavior in *Drosophila*, as well as apparent motion in human observers. Agrochao et al. (2020) reported analogous behavioral responses to static motion-illusion patterns in flies and humans and implicated direction-selective T4/T5 neurons in the fly response. This behavioral result makes the fly visual system a relevant setting for asking how motion signals can arise from spatial structure without physical image motion. This motivates a shared question about visual computation, not a claim of identical percepts or mechanisms across species.

The *Drosophila* pathway links computation to identified circuitry: T4/T5 provide direction-selective ON/OFF signals, and columnar cell types and synapses are mapped (Maisak et al. 2013; Strother et al. 2017). Flies and humans can share motion-estimation strategies (Clark et al. 2014), while ON/OFF asymmetries relate circuit responses to natural-scene velocity estimates (Fitzgerald et al. 2011; Leonhardt et al. 2016). This makes the fly pathway a tractable setting for studying static-pattern responses.

FlyVis provides a model in which this question can be asked with the visual connectome held fixed. The recurrent network maps onto identified fly visual cell types and is trained end-to-end for optic-flow estimation from rendered Sintel videos, following the original FlyVis task rather than introducing a new training objective here (Lappalainen et al. 2024). Sintel is a computer-generated movie dataset, not a sample of a fly’s natural visual experience. The task supplies a computational objective for a motion pathway; it does not make the model’s output a biological behavior. Training does not use neural-response supervision, yet the ensemble median correctly predicted measured ON/OFF preferences for all 32 cell types with established preferences; the best-performing individual model matched 30/32. The networks also recovered ON-motion preference in T4, OFF-motion preference in T5, and the lack of direction selectivity reported for nine T4/T5 input types (Mi1, Tm3, Mi4, Mi9, Tm1, Tm2, Tm4, Tm9, and CT1). Predictions were not uniformly

correct, including a reversed Tm4 response (Lappalainen et al. 2024). FlyVis is therefore an empirically compared, constrained model, not a complete account of fly vision.

In the fly, local T4/T5 motion signals feed downstream wide-field circuits, including lobula-plate tangential cells such as HS and VS, which integrate optic-flow patterns for self-motion and optomotor responses (Borst and Egelhaaf 1989; Lappalainen et al. 2024). FlyVis replaces this downstream biological readout with a learned two-layer convolutional decoder that maps recurrent activity to a dense optical-flow field. The decoder receives instantaneous activity, while temporal integration and motion-sensitive dynamics arise in the network; it is a readout of network computation, not a stand-alone motion detector. The readout can shape measured output. Decoded flow is therefore a model-level measure for comparing static-pattern responses across network states, learned solutions, and readout pairings, not fly perception or behavior.

A static-pattern response would connect fly behavior for static patterns with task learning in a connectome-constrained motion network. It also raises how independently optimized models with the same wiring can arrive at different solutions (Lappalainen et al. 2024). MIDD varies local luminance sequences and their spatial combination. Human psychophysics related responses to four-level patterns to component three-level patterns (Kobayashi and Watanabe 2023); we test a related scalar composition without equating human and model measurements. Static-illusion output and variability across trained instances have also been reported in a video-prediction network (Watanabe et al. 2018; Kirubeswaran and Storrs 2023).

We ask whether static MIDD evokes directional output in optic-flow-trained FlyVis, how that output develops during learning, and what computations could produce it. We distinguish the Original/Mirror-dependent component from output shared across the pair, test local-to-global scalar composition, and compare FlyVis with a classical nonlinear motion-detector array as a computational reference. We also ask why the same output-level relations have different expressions across models, phases, and renderings, including how the trained network–decoder pair shapes the readout. This design treats decoded flow as a model output and does not equate it with perception or behavior.

Results

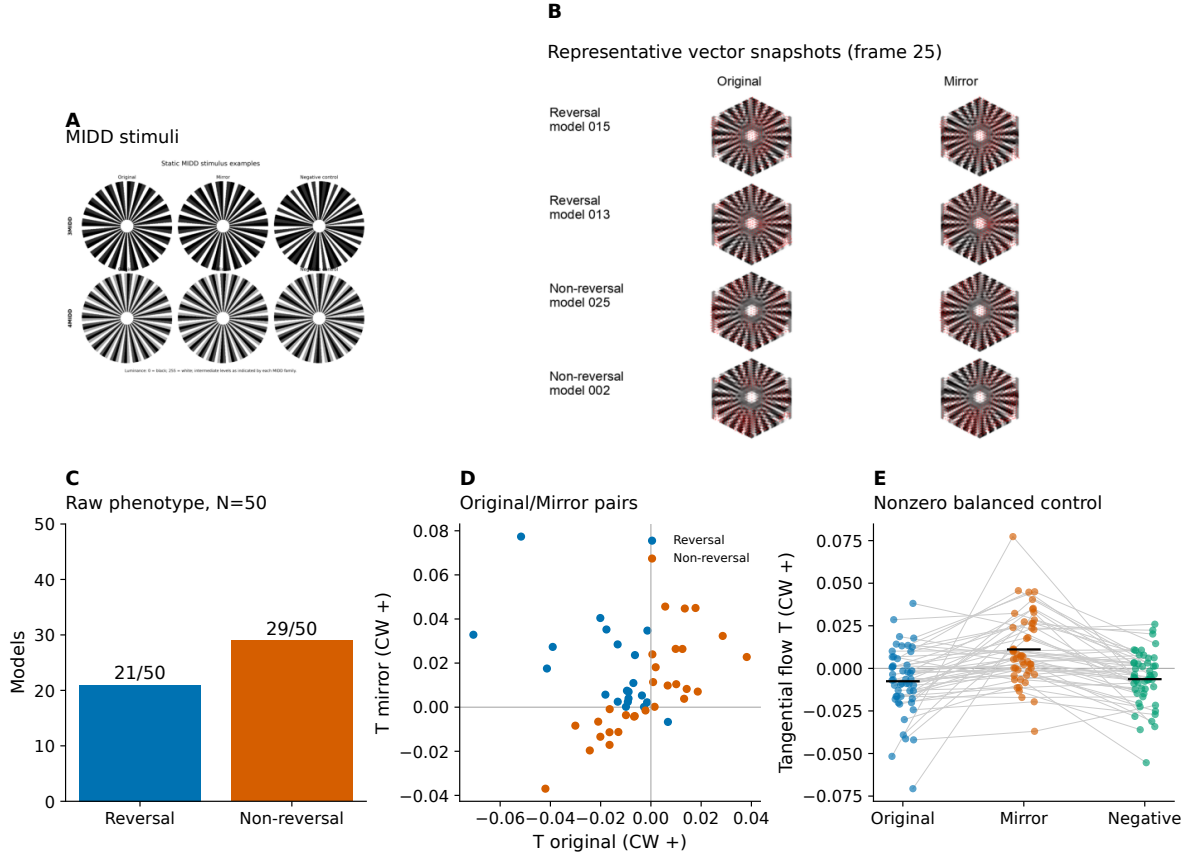
Static patterns evoke decoded motion across FlyVis models

We asked whether static MIDD patterns produce directional output in optic-flow-trained networks. Repeated 3MIDD and 4MIDD Original/Mirror frames generated organized decoded vector fields across 50 official models (Fig. 1). Raw reversal was defined at phase zero by opposite signs of Original and Mirror mean signed tangential flow ($T_O T_M < 0$); 21/50 models reversed at 3MIDD C=48. Measurable output and reversal are distinct outcomes.

Phase zero is the shared reference for Figs. 1, 2, and 5; Fig. 4 evaluates its phase dependence. Balanced negative patterns also produced measurable output. Physical clockwise and counterclockwise propeller controls gave the expected decoded tangential signs in 50/50 models (Supplementary Fig. S1). In three selected models, decoded output remained measurable in the 9–10 s window of prolonged static presentation (Supplementary Fig. S3).

Output-level decomposition. We define $A = (T_O - T_M)/2$ and $S = (T_O + T_M)/2$ for the Original/Mirror tangential outputs; the raw pair reverses exactly when $|A| > |S|$ (full identities in Methods; Fig. 4). This is an exact relation among decoded scalar outputs.

Figure 1. Static patterns evoke decoded motion



Official pretrained models 000–049 (N=50), native BoxEye, frames 10–49, CW-positive tangential flow. **A**, 3MIDD C=48 and 4MIDD 85/170 Original, Mirror, and balanced negative examples. **B**, vector fields for models 015/013 (raw reversal) and 025/002 (no reversal), frame 25. **C**, raw reversal counts for 3MIDD C=48. **D–E**, paired tangential outputs for Original/Mirror and negative conditions. Physical-direction calibration appears in Supplementary Fig. S1.

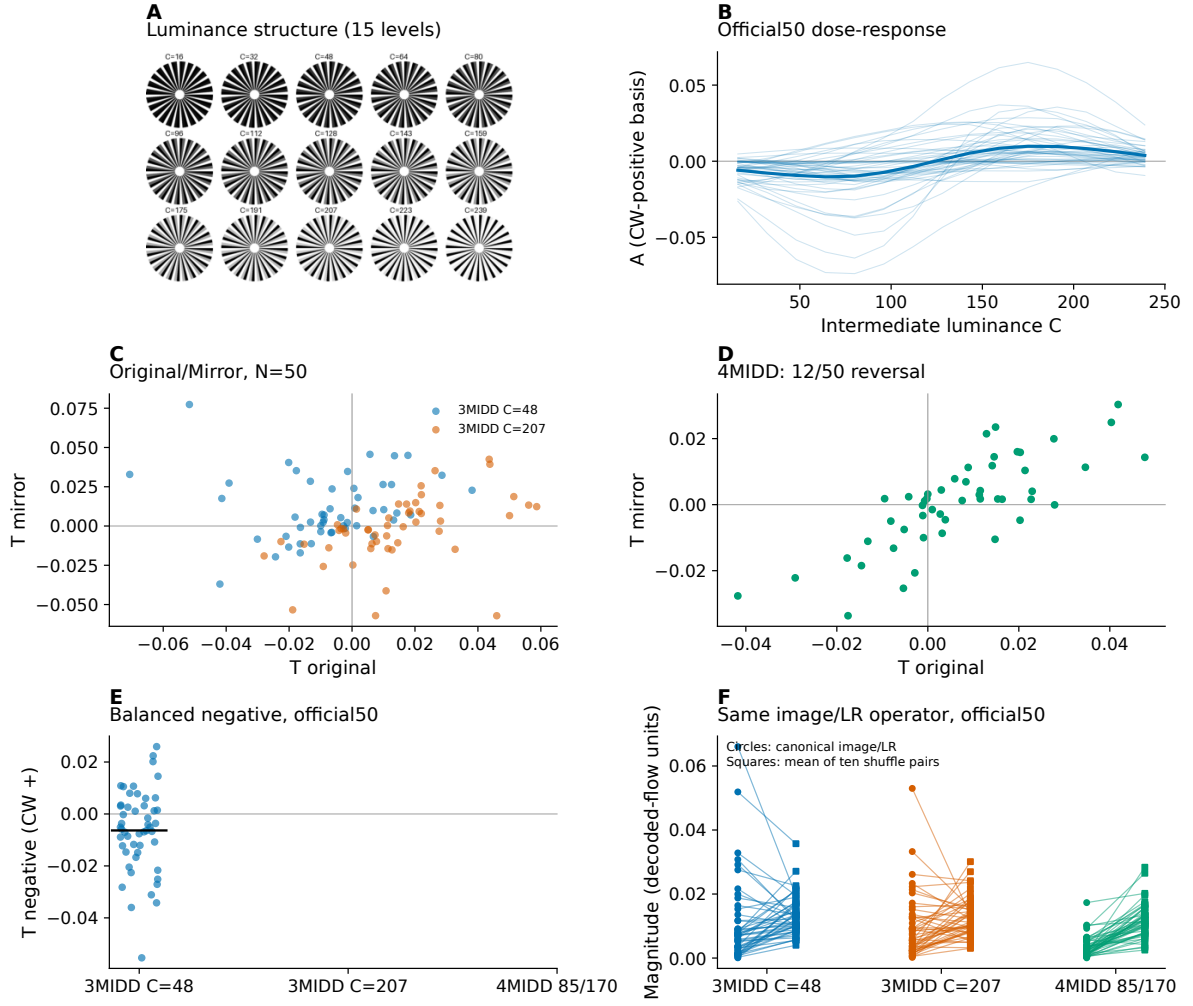
Static decoded motion is not specific to canonical MIDD

We evaluated 15 intermediate luminances in the native BoxEye rendering across the official 50 models (Fig. 2). The ensemble-mean antisymmetric response A was negative from $C=16$ – 112 and positive from $C=128$ – 239 . At the phase-zero reference, mean A was -0.00935 for $C=48$ and $+0.00826$ for $C=207$, with raw reversal in 21/50 and 18/50 models, respectively. For 4MIDD $C1=85/C2=170$, mean A was $+0.00321$ and reversal occurred in 12/50 models. Negative controls had nonzero mean tangential output in all three conditions.

Histogram-preserving sector shuffles produced antisymmetric magnitudes comparable to or larger than MIDD: canonical versus shuffled means were 0.01062/0.01331 ($C=48$), 0.00940/0.01260 ($C=207$), and 0.00340/0.01170 (4MIDD; Fig. 2; Supplementary Fig. S5; Table S2). Uniform gray, histogram-matched random pixels, and balanced negatives also produced measurable output; these tests show no greater response to MIDD than to the tested static controls. In a separate ten-model sweep, canonical 72/96-sector stimuli were not the normalized-response maximum in any model; response varied with spatial frequency, edge density, and sampling (Supplementary Fig. S6).

Static output was also broader than MIDD: across 14 learning seeds, sector-shuffle, balanced-negative, and uniform-input responses changed during optic-flow training (Supplementary Fig. S8). Thus, decoded motion occurred for multiple structured and unstructured inputs, without evidence of MIDD specificity.

Figure 2. Luminance dependence and static controls



Official pretrained models (N=50), native BoxEye, CW-positive tangential flow. **A**, 3MIDD luminance examples. **B**, antisymmetric response A across 15 intermediate luminances. **C–E**, raw Original/Mirror and balanced-negative outputs for C=48, C=207, and 4MIDD 85/170. **F**, paired canonical and mean sector-shuffle magnitudes using the same image/reflection operator. Lines connect measurements from the same model.

Local three-level responses predict global four-level antisymmetric scalar motion

We tested whether the global four-level antisymmetric scalar follows an affine composition law based on independently measured local three-level components. In ten official models and a common centered eight-field rendering, the relationship was

$$A_4(C_1, C_2) \approx \alpha [A_3(C_1) - A_3(C_2)] + \beta.$$

Across seven luminance levels, model-wise fits had mean $R^2 = 0.9295$ and the unfitted component signs agreed with observed signs for 96.5% of evaluable pairs (Fig. 3). The mean fitted slope was 0.8292 and the intercept was 0.00737, so this is an affine composition law rather than identity or exact linear summation.

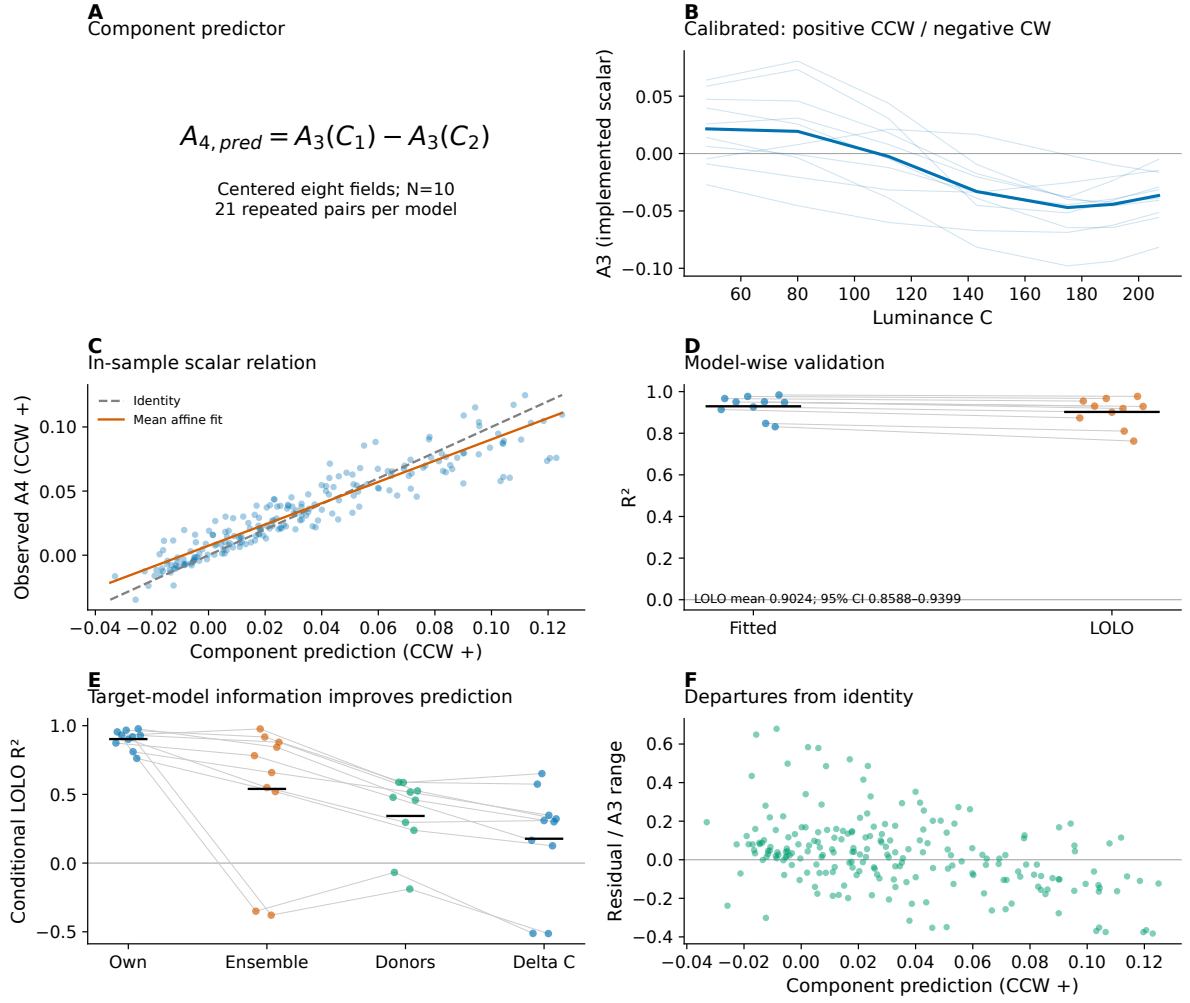
Conditional leave-one-luminance-out mean held-out R^2 was 0.9024 (Supplementary Fig. S7; Table S6). Because the A3 value at the held luminance remained measured, this is conditional prediction from local component responses, not prediction without those measurements. Physical CW/CCW calibration gave negative scalar output for CW and positive output for CCW in all ten models. The result is specific to the centered eight-field rendering and scalar projection; it does not establish full-vector-field additivity. It resembles the compositional form reported in human MIDD psychophysics, but the model and human responses were not directly compared (Kobayashi and Watanabe 2023).

The strength of this scalar relationship depended on rendering. In a separate seven-field comparison, fitted mean R^2 was 0.439 with BoxEye and 0.934 with `highres_232`, with sign agreement of 0.458 and 0.975, respectively (Supplementary Fig. S2). The strong affine composition is therefore established for the centered eight-field condition; transfer to other rendering geometries is limited. This scalar result does not establish additivity of the full vector field.

We compared the FlyVis output with a classical Reichardt-detector array as a computational reference (Supplementary Fig. S4). Summing the local detector outputs linearly yielded near-zero directional output for static 3MIDD and 4MIDD. Applying the sign-preserving saturation used in the Bach-type array to each local detector before spatial summation instead produced nonzero static directional residuals (Bach and Atala-Gérard 2020). After calibrating the model’s sign against physically rotating stimuli, the array and FlyVis had the same net tangential direction and the same response-magnitude ordering (3MIDD greater than 4MIDD) for the canonical patterns.

The calibrated sign comparison concerns raw tangential output, not the mirror-antisymmetric component A . The agreement was partial. In a separate 21×21 4MIDD luminance landscape for FlyVis model 000, the mirror-antisymmetric directional components showed 81.0% descriptive sign agreement over robust grid points and Pearson correlation $r = 0.455$. These grid values are a descriptive census, not independent trials or an inferential test. The FlyVis mirror-symmetric component had 3.14 times the RMS of its directional component. Near the canonical 4MIDD setting, the directional component itself had opposite signs in the two models; the larger positive FlyVis shared component determined the positive raw output. Thus the calibrated raw-output agreement does not imply that the models share the full directional component. The classical array demonstrates that standard nonlinear motion-detector computation can generate part of the directional structure, but it does not explain the full FlyVis response or identify the nonlinearity that generates it in FlyVis. Follow-up seed and mechanism analyses also did not establish a single upstream nonlinear residual as a universal explanation.

Figure 3. Local-to-global scalar prediction



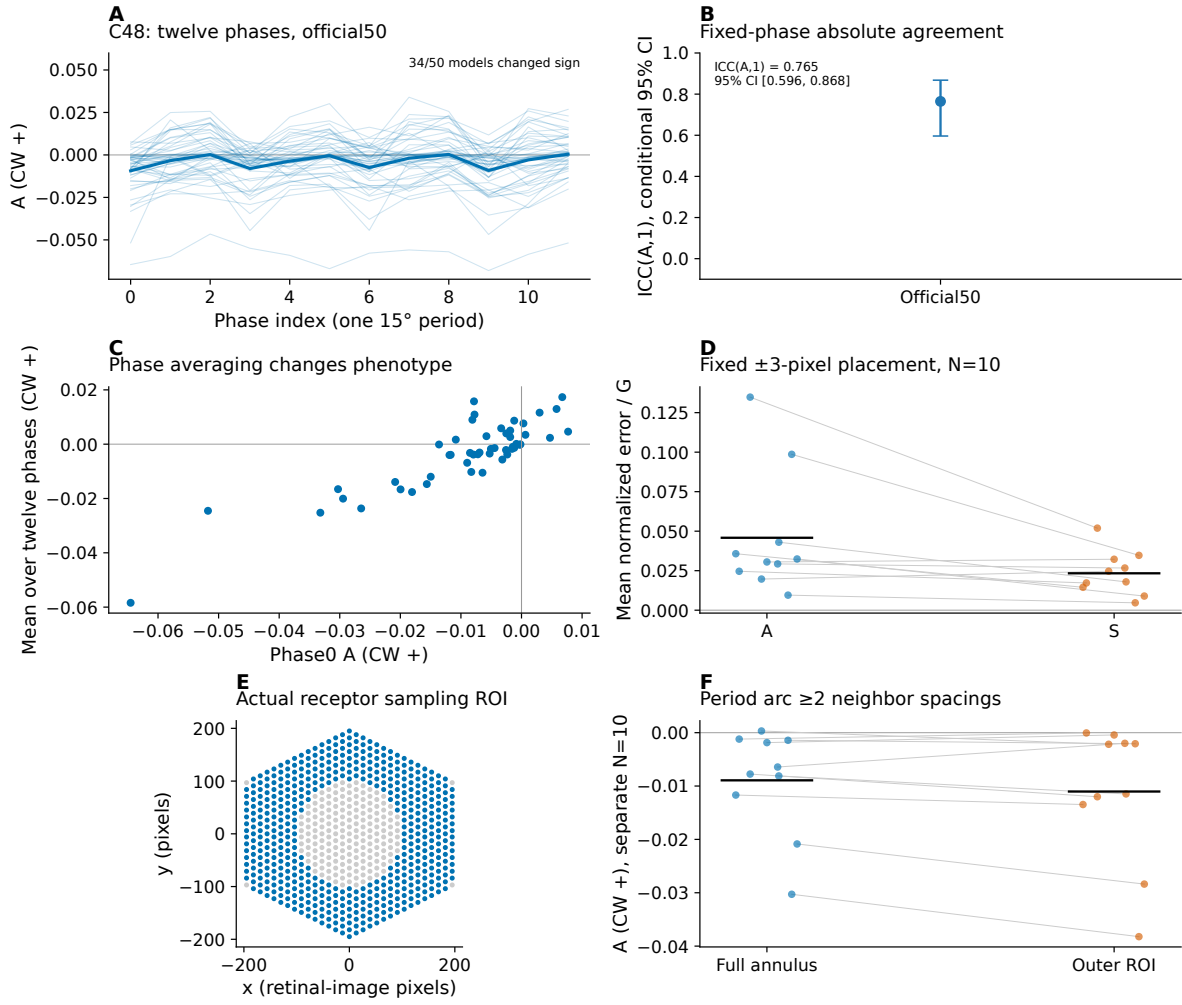
Official models 000–009 (N=10), centered (3,2,3) eight-field rendering, seven luminances and 21 pairs per model. **A–C**, local component responses and observed versus predicted global scalar, with identity and mean affine-fit lines. **D–E**, fitted and leave-one-luminance-out model R² for the listed predictors. **F**, identity residual normalized by the A3 range. Points represent model-level summaries; intervals are shown where indicated.

Response sign and magnitude vary across models and stimulus phase

We quantified phase dependence for C=48 across the 50 official models at twelve phases of the 15° color-unit period (Fig. 4). Absolute-agreement ICC(A,1) across phases was 0.7647, while 34/50 models changed the sign of A across phases. Thus, phase consistency at the ensemble level coexists with frequent within-model sign changes. At phase zero, mean A was -0.00935 ; averaging across phases gave -0.00378 . The phase-zero and phase-averaged model signs agreed in 39/50 models (Table S6).

Thus, models with the same connectome differ in decoded-output sign and magnitude, which can also change with stimulus alignment. Phase-zero values in Figs. 1, 2, and 5 are reference-condition results. Additional placement and receptor-sampling analyses also changed the measured A and S components, supporting a role for rendering geometry in the measured response (Supplementary Figs. S5, S6).

Figure 4. Response variability across phase and geometry



A–C, official models 000–049 (N=50), C=48 A across twelve phases, phase reliability, and phase-zero versus phase-mean responses. **D**, placement sensitivity in models 000–009 (N=10). **E–F**, receptor locations and full versus outer analysis regions for the separate ten-model analysis. Tangential flow is CW-positive.

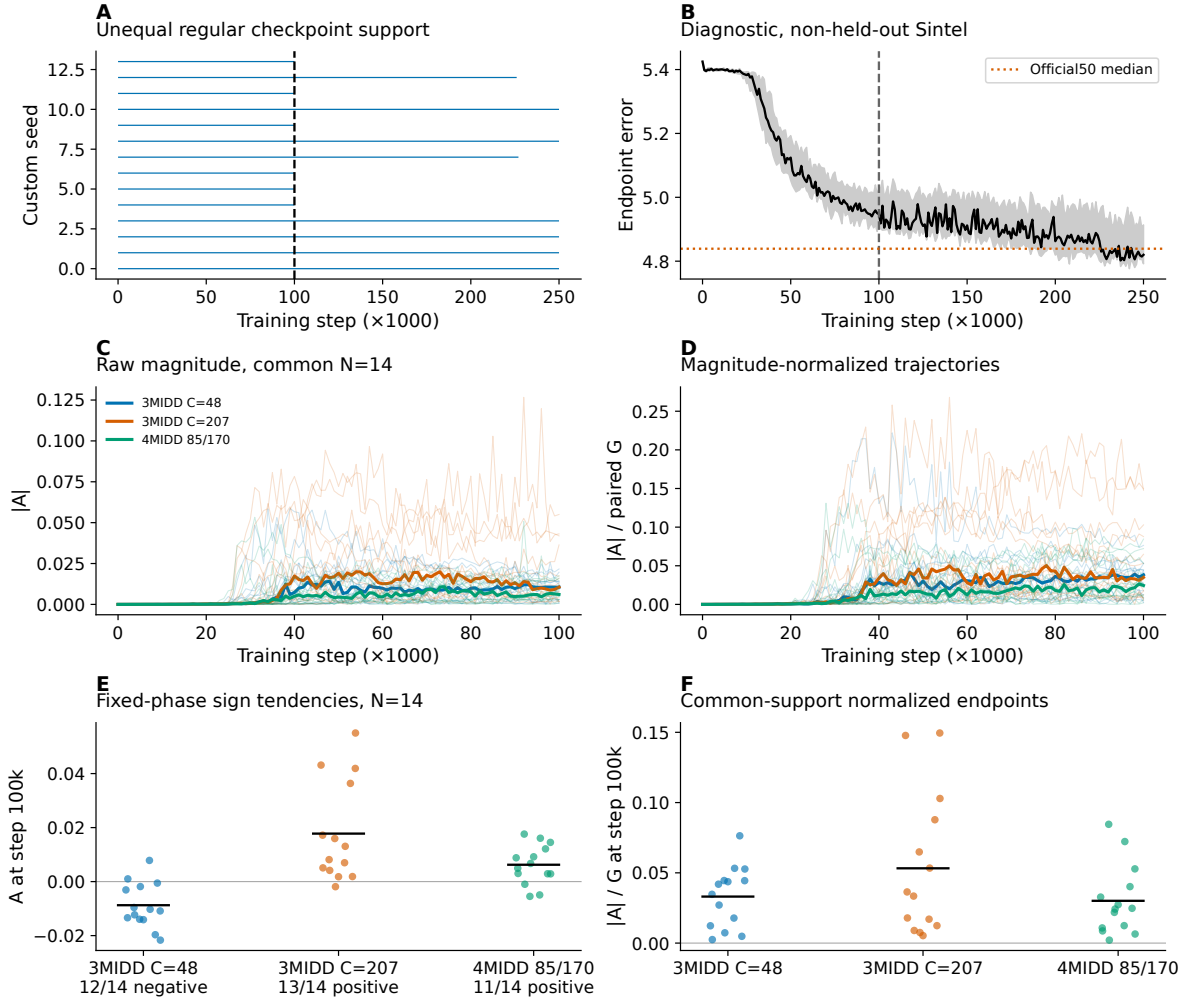
Static-pattern responses emerge during optic-flow learning

We followed 14 independently trained learning-custom seeds from initialization through common training checkpoints (Fig. 5). At initialization, $|A|$ was below 3×10^{-9} for every seed and stimulus family. During ordinary optic-flow training, antisymmetric magnitude increased; at step 100,000, the median $|A|$ normalized by paired mean flow magnitude was 0.03832 for C=48, 0.03493 for C=207, and 0.02449 for 4MIDD. MIDD was evaluated after training and was not a training target.

At 100,000 steps, A was negative in 12/14 seeds for C=48, positive in 13/14 for C=207, and positive in 11/14 for 4MIDD. The two-sided exact binomial p values were 0.0129, 0.00183, and 0.0574; Bonferroni-adjusted values across these three conditions were 0.0388, 0.0055, and 0.1721, respectively (Table S6). The majority direction was maintained at every checkpoint from 90,000 through 100,000 steps: 11–12/14 seeds were negative for C=48, 13/14 were positive for C=207, and 11–12/14 were positive for 4MIDD. Thus, the sign tendency was robust over this late checkpoint window for C=48 and C=207, but not statistically supported for 4MIDD after correction. Other static controls also developed measurable responses during the same training interval (Supplementary Fig. S8). The effect on the raw Original/Mirror pair depended on the joint decoded values of A and S ; these seed-level results describe output variation and do not identify differences in internal implementation. Sintel EPE improved during training, but task performance alone did not uniquely specify this static-pattern phenotype. Responses to other controls likewise show that this emergence was not specific to MIDD.

Across the official 50 models, Sintel endpoint error had a Spearman association of $\rho = 0.284$ with signed A (95% bootstrap CI $[-0.003, 0.534]$, raw $p = 0.0458$, Holm-adjusted $p = 0.0915$), $\rho = -0.215$ with $|A|$ (95% CI $[-0.502, 0.096]$, $p = 0.134$, adjusted $p = 0.134$), and $\rho = -0.326$ with phase variability, measured as $SD(A)$ across phases divided by mean paired flow magnitude (95% CI $[-0.586, -0.023]$, $p = 0.0208$, adjusted $p = 0.0624$). None of the three associations remained below 0.05 after Holm correction; these data do not establish a reliable relation between task error and the decoded MIDD phenotype (Table S2).

Figure 5. Static-pattern response during optic-flow learning



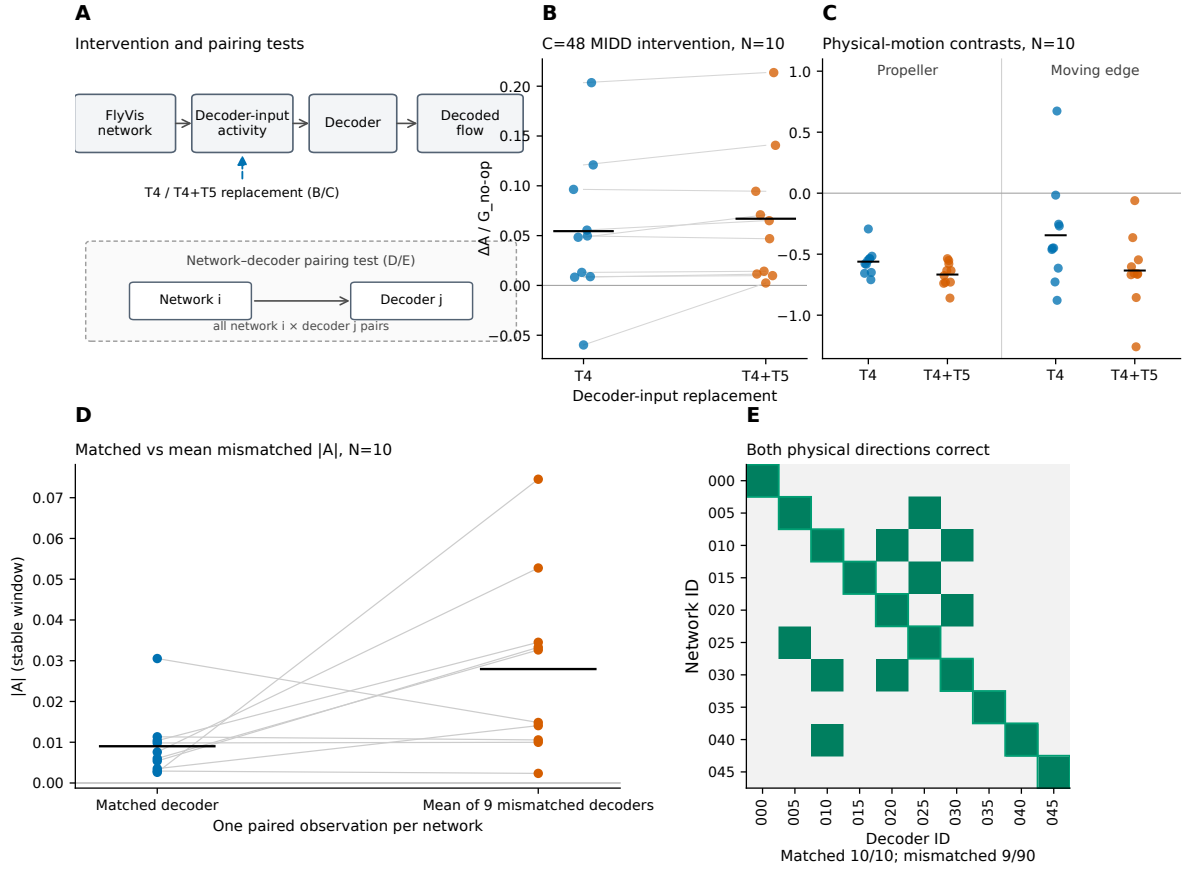
Learning-custom seeds 000–013 ($N=14$). **A**, checkpoint support. **B**, diagnostic Sintel EPE; the dotted horizontal line marks the median fixed-evaluation Sintel EPE of the official 50-model cohort. **C–D**, absolute A and A normalized by paired mean flow magnitude over common training support. **E–F**, signed A and normalized endpoint values at 100,000 steps. Thin lines and points denote seeds; summary bars show seed medians and interquartile ranges.

Broad interventions and network–decoder pairing constrain decoded output

We compared decoder-input replacement across static MIDD and physical-motion stimuli in official models 000–009 (Fig. 6B–C; Supplementary Fig. S9). Targeted channels were replaced by frame-matched uniform-gray activity while recurrent and non-target activity remained intact. The primary endpoint was the signed intervention-minus-no-op contrast normalized by the paired no-op mean flow magnitude. For C=48 MIDD, mean normalized changes were +0.05449 for T4 replacement (95% model-bootstrap CI [0.01361, 0.10001]) and +0.06696 for T4+T5 replacement (95% CI [0.03052, 0.11040]). The same endpoint showed changes for both propeller and moving-edge contrasts in the same cohort. Because the operation affected physical-motion as well as static-pattern output, these results are consistent with a contribution from general motion-processing/readout machinery rather than a MIDD-selective operation; they do not identify a cell-specific causal mechanism.

In a separate 10×10 cross-swap, each of ten selected official networks (000, 005, 010, 015, 020, 025, 030, 035, 040, 045) was paired with each selected decoder. Over stable frames 10–49, mean matched $|A|$ was 0.00902, compared with 0.02796 for the within-network mean across nine mismatched decoders (Fig. 6D; Supplementary Fig. S10). Both rightward and leftward physical directions were decoded correctly for 10/10 matched pairs but only 9/90 mismatched pairs (Fig. 6E; Supplementary Fig. S10; Tables S3 and S6). The 90 mismatched combinations are structured pairings, not independent models, and their values are summarized within network. Because mismatched pairs also generally failed ordinary physical-motion readout, the larger mismatched $|A|$ cannot be interpreted as a MIDD-specific pairing effect. The results show that decoded magnitude and physical-direction readout depend on network–decoder pairing, consistent with a co-adapted readout, but they do not causally decompose network, decoder, and interaction contributions.

Figure 6. Circuit and readout constraints on static-pattern decoded motion.



A, noncausal schematic of the decoder-input replacement and network–decoder pairing analyses. **B**, C=48 MIDD in official models 000–009 (N=10): signed intervention-minus-no-op change normalized by the paired no-op mean flow magnitude for T4 and T4+T5 replacement. Points denote models, light lines connect paired observations, and black bars show group means. **C**, the same normalized signed endpoint for propeller and moving-edge physical-motion contrasts in the same cohort; points denote models and black bars show group means. **D**, stable-window C=48 $|A|$ for the selected official models 000, 005, 010, 015, 020, 025, 030, 035, 040, and 045 (N=10): matched decoders versus the within-network mean across nine mismatched decoders. Each point is one network; lines connect its paired observations, and black bars show group means. **E**, the same 10×10 network-by-decoder cohort shown as a correctness matrix for both rightward and leftward physical directions; both directions were correct for 10/10 matched pairs and 9/90 mismatched pairs.

Discussion

Static MIDD patterns evoked directional decoded flow in optic-flow-trained FlyVis networks, but the response was not specific to canonical MIDD: sector shuffles and other static controls produced comparable or larger outputs. The findings therefore concern how task-trained motion networks and their readouts respond to static structure, not a dedicated illusion response.

This output-level decomposition summarizes the measured pair without identifying separate neural pathways. Under the centered eight-field rendering, measured local three-level responses conditionally predicted held-out four-level scalar outputs (mean $R^2 = 0.9024$). This affine prediction has a non-unit slope and depends on rendering; the separate seven-field BoxEye comparison was weaker. The result is a conditional scalar prediction, not a full-vector identity.

Candidate computational account

One candidate account is that nonlinear processing leaves local directional residuals that do not cancel during spatial aggregation. The classical Reichardt array produced such static residuals, and its calibrated raw tangential direction agreed with FlyVis for the canonical 3MIDD and 4MIDD conditions. The correspondence was partial: near canonical 4MIDD, the mirror-antisymmetric components had opposite signs, and a one-model luminance landscape showed descriptive rather than inferential agreement. This comparison establishes a computational possibility, not the causal mechanism in FlyVis.

Learning and output variability

The antisymmetric component emerged from near zero during ordinary optic-flow learning, although MIDD was not a training target. Across official models, absolute-agreement ICC(A,1) across phases was 0.7647, while 34/50 changed the sign of A with phase. In learning seeds, the majority sign tendencies at $C=48$ and $C=207$ persisted across the 90,000–100,000-step checkpoints; the 4MIDD sign count was weaker after correction. These are differences in decoded output across model, phase, and seed. They do not show that internal implementations differ across seeds. The three official-cohort associations with Sintel endpoint error did not survive Holm correction, so task performance did not reliably account for the measured output variation. For signed A , the estimate was Spearman $\rho = 0.284$ (95% CI $[-0.003, 0.534]$), and phase variability was associated at $\rho = -0.326$; with $N=50$, these nonsignificant corrected tests do not exclude moderate associations.

Network–decoder readout and biological context

Cross-swaps and broad decoder-input replacements show that the decoded magnitude and direction readout depend on network–decoder pairing and that T4/T5 input replacement also changes physical-motion output. Because only 9/90 mismatched pairs decoded both physical directions correctly, the cross-swap result cannot isolate a MIDD-specific pairing effect from broader readout failure. Agrochao et al. linked fly behavior to T4/T5 imbalance (Agrochao et al. 2020); the present interventions provide a qualitative circuit-level point of contact, but replace broad decoder-input populations rather than reproducing that neuronal perturbation or measuring behavior. Saved T4c direction-selectivity labels showed a suggestive association with phase-averaged 3MIDD $C=48$ A (permutation $p = 0.0176$; Holm-adjusted $p = 0.0528$), but the corrected result was inconclusive and does not establish T4c causality (Table S5A). In the fly, T4/T5 signals are integrated by downstream wide-field circuits such as lobula-plate tangential cells, whereas FlyVis maps network activity to a dense flow field with an artificial decoder.

Limitations

Decoded flow is a model readout, not a direct measurement of fly or human perception. We did not compare it with behavior or conduct a matched quantitative human-psychophysics experiment. Because we did not compare against an unconstrained recurrent network, these results do not establish that the responses are specific to connectome constraints. The local-to-global result is limited to an affine scalar prediction under a centered eight-field rendering and conditional held-out-luminance evaluation. Decoder-input replacements are model interventions, not biological knockouts, and their effects on physical-motion outputs limit claims of MIDD selectivity. The classical detector comparison is a partial computational analogy; it does not identify a causal nonlinear stage in FlyVis. Future work can test candidate computations while preserving ordinary motion readout and compare model outputs with behavior under matched stimulus conditions.

Methods

FlyVis model and model cohorts

We used connectome-constrained recurrent FlyVis networks and their learned optical-flow decoders (Lappalainen et al. 2024). The reproducibility environment uses FlyVis v1.2.0, commit 92b3845cc426dd309a1a0e1b3890156c42e14021. Official pretrained data are obtained with `flyvis download -pretrained`; FLYVIS_ROOT_DIR locates upstream models, results, and rendering caches separately from this repository’s stimuli, precomputed analysis products, and custom checkpoints.

The official `flow/0000/000-049` ensemble supplies Figs. 1–2, the primary C=48 phase analysis in Fig. 4, expanded static/task-performance controls, and native S1 calibration. Fig. 3, Fig. 6 decoder-input replacement, and Supplementary Fig. S9 use experiment-specific ten-model subsets (000–009). Fig. 5 uses learning-custom seeds 000–013. Recurrent subtype interventions use established seeds 001, 002, 003, 010 (002 holdout) and comparators 000, 008 at step 250,000. Cross-swap uses official models 000, 005, 010, 015, 020, 025, 030, 035, 040, 045. Matching numeric IDs do not imply matching models across sets. Cohorts and analysis-specific measurement conditions are summarized in Table S1.

MIDD stimuli and rendering

For the single-BoxEye stimuli, `src/flyvis_midd/stimuli.py` constructs 3MIDD Original sectors by repeating `[0,C,255]` 24 times and Mirror sectors by repeating `[255,C,0]`. The balanced negative repeats `[0,C,0,255,C,255]` 12 times. Each pattern therefore contains 72 equal-angle sectors. The 4MIDD Original repeats `[0,C1,255,C2]` 24 times; Mirror repeats `[0,C2,255,C1]`; the negative repeats `[0,C1,255,C2,0,C2,255,C1]` 12 times, yielding 96 sectors. Mirror denotes the specified change in spatial luminance order, not contrast inversion. The negative arrangements balance directed transitions and are analyzed as distinct stimuli.

The renderer uses 382×382-pixel images, inner/outer annulus diameters of 55/382 pixels, phase 0°, and white pixels outside the stimulus annulus. Integer luminances are divided by 255 before model input. The fixed native reference conditions are 3MIDD C=48 and C=207 and 4MIDD C1=85/C2=170. Fig. 2 uses the 15 C levels 16,32,48,64,80,96,112,128,143,159,175,191,207,223,239 plus that fixed 4MIDD condition. Static movies repeat one image for 50 frames.

Single-BoxEye movies use `BoxEye(extent=15, kernel_size=13)` (721 receptors; shape (50,1,721)). Fig. 3 uses a centered (3,2,3) eight-field layout (decoded shape (8,50,2,721)) and luminances 48,80,112,143,175,191,207; S9 activity uses the same layout, while replacement analyses include BoxEye and `highres_232`, with results kept rendering-specific. For Fig. 3, the image is centered

on a black 3:2 canvas with horizontal padding and row offsets [0,1,0]. The high-resolution helper fits and bilinearly resizes the canvas (`align_corners=False`, rounded dimensions); placement is right-aligned at the layout’s upper edge. Receptor values average 13×13 patches with zero padding. Global aggregation rounds (y,x) coordinates to eight decimals and averages coincident vectors per frame, negates v once, and averages tangent projection ($dy/r, -dx/r$) over unique positions and frames 10–49 within the 0.14–0.98 image-radius annulus centered on the placed-image bounds. This multi-field basis is opposite to the single-BoxEye clockwise basis ($-dy/r, dx/r$); eight-field calibration yields positive CCW and negative CW scalars in all ten models and does not automatically transfer to the seven-field S2 layout.

Stimulus presentation and decoded optical flow

Static movies use 50 Hz ($dt=0.02$ s), 50 frames, and 1-s fade-in/state preparation. Figs. 1–2 use `run_flow_inference`: CPU float32 luminance, `network/decoder.eval()`, `torch.no_grad()`, `network.fade_in_state(1.0,dt,luminance[:,0])`, then `network.simulate(luminance,dt,initial_state=initial_state)` and decoder flow. Fade-in frames are excluded from the recorded movie. `analyze_midd_flow` measures canonicalized output at frames 10–49; `aggregate_midd_metrics` computes A/S/M and raw reversal, with negative responses separate. Expanded official controls and phase runs use the same inference; composition, learning, interventions, and cross-swap use their specified preparations and saved outputs.

Decoder flow is canonicalized to (`sample,frame,2,receptor`) with components (u,v); the zero-based analysis window is frames 10–49. Fig. 1 representatives are the reversal/non-reversal models nearest each group’s median $(\text{abs}(T_0) + \text{abs}(T_M))/2$; vector snapshots use frame 25. This descriptive selection defines no weak-response class.

The S1 control uses static, clockwise, and counterclockwise three-blade propeller movies with 50 frames at 50 Hz and 128×128 -pixel images. Direction agreement requires positive clockwise and negative counterclockwise `signed_tangent_ratio`. Rotation detection additionally requires the smaller moving-condition absolute ratio to exceed 0.1 and the smaller moving mean-flow magnitude divided by the static mean-flow magnitude to exceed 3.0. Direction agreement and detection are separate endpoints; propeller Static is not the MIDD negative condition.

Tangential rotation and Original/Mirror decomposition

Main single-BoxEye analyses use `image_rotation_flow_metrics` in `src/flyvis_midd/rotation.py`. Receptor centers are in image coordinates, with x rightward and y downward. Decoder vectors are converted exactly once as $(u,v) \rightarrow (u,-v)$ before projection. For receptor displacement (dx_i, dy_i) from the mean receptor center and radius r_i , the clockwise tangent is $(-dy_i/r_i, dx_i/r_i)$. The analysis selects $0.14 \leq r_i \leq 0.98$; the official single-BoxEye tables contain 698 selected receptors. `mean_signed_tangent`, denoted T , is the arithmetic mean of vector–tangent dot products over selected receptors and analyzed frames. Positive T denotes clockwise rotation. Values are decoded-flow units, not psychophysical velocities.

`signed_tangent_ratio` divides T by the mean Euclidean flow magnitude in the same window/annulus, using machine epsilon as a denominator floor. It is used for S1 detection and is not substituted for T in the main A/S analyses. Precomputed scalars already use the coordinate conversion defined for each experiment and are not flipped again. The multi-field global scalar and the activity/replacement analyses’ image-radius ROI use their experiment-specific definitions rather than the single-BoxEye definition.

Within each model, condition, and checkpoint, we compute $A=(T_O-T_M)/2$, $S=(T_O+T_M)/2$, and $M=|A|-|S|$. Actual reversal uses the strict raw sign predicate $T_O-T_M<0$, which is algebraically equivalent to $M>0$. Negative responses do not enter these definitions. Subtracting a common T_N preserves A but alters S and the raw reversal measure. Official-50 component distributions provide descriptive context in Supplementary Fig. S5; Fig. 4 combines the primary official-50 C=48 phase analysis with separate ten-model placement and radial analyses.

Scalar composition and learning-dynamics analyses

For Fig. 3, A3(C) and observed A4 are Original/Mirror half-differences; the predictor is $A3(C1)-A3(C2)$. Each model has an intercept OLS fit over 21 canonical C1<C2 pairs, with $R^2 = 1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$. Identity residuals are observed A4 minus the unfitted predictor; RMSE is normalized by the A3 range. Metrics are averaged equally across models, not pooled across 210 pairs. Sign accuracy uses the unfitted predictor for evaluable pairs (18–21/model), defined by $|A3(C1)-A3(C2)|/\text{range}(A3) \geq 0.05$. This scalar analysis does not test full-vector additivity.

For Fig. 5, the checkpoint archive contains 2,586 files: 2,567 regular checkpoints, including initialization at step 0, and 19 irregular extras excluded from the trajectories. Regular support comprises step 0 and available 1,000-step intervals through each seed’s endpoint. A/S/M are computed from paired Original/Mirror responses at each checkpoint. Operational onset is evaluated over fixed thresholds 0.0005, 0.001, 0.002, and 0.005 using both first crossing and the first of three consecutive regular checkpoints above threshold. Gaps are not bridged. Noncrossers are censored, and crossing medians explicitly describe crossers. Primary summaries use common steps 0–100,000. We additionally compute AUC and time to a fixed fraction of final $|A|$ as retrospective, threshold-free descriptors. Normalized magnitude is $|A|/G$, with G the mean Original/Mirror Euclidean flow magnitude in the same annulus/window; zero denominators are flagged. At each step, medians and interquartile ranges use only seeds with data at that step; checkpoint observations are repeated measurements within seeds.

Fixed-evaluation Sintel EPE uses 15 samples from sequence IDs 0,5,10,15,22, no augmentation or random temporal cropping, 19 source frames resampled/interpolated to 40 evaluated frames at $\text{dt}=0.02$ s, and a center-crop fraction of 0.7. This diagnostic evaluation is not held out and is distinct from training loss. The same regular checkpoint support is required for MIDD and EPE tables. Official-50 evaluation uses MultiTaskSintel with three vertical splits of original sequence IDs 0,5,10,15,22 (15 samples), `all_frames=False` and resampling/interpolation enabled. Model/decoder evaluation mode and `no_grad` are used after a 0.5-s steady-state preparation. Batches of four (last batch three) are weighted by sample count; EPE is the mean vector endpoint error over all 40 resampled frames and 721 receptors. Checkpoint hashes were verified against source model identities and canonical Original/Mirror responses checked before accepting EPE. The three official-50 EPE associations use $A(\text{C48, phase } 0)$, its absolute value, and $\text{SD}(A \text{ across phases})/\text{mean paired } G$; 10,000 whole-model bootstrap intervals and Holm correction across these relations are reported.

Cell activity and interventions

The decoder-input activity analysis examines activity from 34 decoder-input cell types in ten models. Moving-edge calibration specifies 12 directions at 30° intervals, intensities 0 and 1, speed 19, $\text{dt}=0.005$ s, offsets $[-10, 11]$, height 80, and 1 s pre/post periods. The primary preferred direction is leave-one-model-out (LOMO). MIDD cell evidence is the projected Original/Mirror half-difference, normalized and screened for identifiability as defined below. Signed and absolute Spearman cou-

pling are evaluated within models across the seven C levels. Supplementary Fig. S9 shows these identifiable activity entries; available model counts vary by subtype.

Decoder-input replacement substitutes a frame-matched baseline for targeted channels without re-simulating the recurrent network. The corrected-family endpoint is normalized loss $L = 1 - \text{gain}$; T4a is primary, T4b exploratory, and T4/T5 groups are defined separately. No-op and nontarget-invariance checks assess specificity; alternate estimates are not pooled. For the decoder-input activity analysis, raw evidence $e(C)$ is the space/time mean of the Original/Mirror activity half-difference multiplied by the LOMO preferred-direction tangent weight within the common annulus. Across seven C levels, $R = \sqrt{\text{mean}_C e(C)^2}$, normalized evidence is $e(C)/(R + 10^{-8})$, and identifiability requires $R > \max_C |e_{\text{negative}}(C)| + 10^{-8}$. For the decoder-input replacement analysis, the baseline averages intact activity over all 21 3MIDD conditions (seven luminances $\times$ Original/Mirror/negative), preserving frame, spatial sample, cell channel, and receptor position. Only targeted channels are replaced. Given intact and intervention seven-luminance A3 vectors x and y , $\text{gain} = (y \cdot x)/(x \cdot x)$, without an intercept; $\text{loss} = 1 - \text{gain}$ and shape residual $= \|y - \text{gain } x\|/\|x\|$. Gain is undefined if $x \cdot x$ is nonfinite or $\leq 10^{-12}$. Bootstrap intervals are computed by resampling model losses with replacement 50,000 times, using random seed 10012 and percentile 2.5/97.5 intervals for the mean. The one-sided sign-flip test enumerates all 2^n sign assignments and uses $p = (1 + \text{number of permuted means} \geq \text{observed mean})/(2^n + 1)$. T4a baseline is a singleton confirmatory family ($q=p$); TmY14, T2a, and Tm16 baseline form the secondary family; group interventions form a separate family; remaining single-type baseline interventions form the exploratory screen. Other interventions retain their catalog families. Benjamini–Hochberg adjustment is applied separately within each nonsingleton family. Separately assigned targets are not included in the exploratory family’s multiple-comparison correction.

Recurrent interventions neutralize T4c/T4d subtypes and compare matched-random node sets in the same instances (199 draws). MIDD summaries average twelve phases (0° – 13.75° , 1.25° steps); paired Sintel outcomes are intervention-minus-intact EPE. The state hook zeros selected activity during MIDD fade-in/simulation and evaluated Sintel frames (after 0.5-s gray preparation). Each random set samples 721 T4 nodes without replacement within twelve sectors, matching target subtype counts; all T4 subtypes, including the target, are eligible. Seed/target/draw sets are reused across phases and Sintel samples. Sintel uses evaluation-role rows of `part4_sintel_scene_disjoint_split.csv` (indices 36,39,40,41,48,49,50; original sequences 12,13,16), excluding calibration sequences 2,7,9. MultiTaskSintel uses three vertical splits, 19 frames, 0.7 center crop, $\text{dt}=0.02$ s, resampling/interpolation, and no augmentation or random temporal crop. EPE differences are averaged across splits within sequence, then equally across sequences per seed; random controls retain draw identity. These are model perturbations, not biological knockouts.

Cross-swap is a descriptive 10×10 network–decoder design for 3MIDD $C=48$ (1-s gray preparation; 50 frames; stable window 10–49). Matched pairs share IDs; each network has nine mismatched decoders. A fixed balanced two-way centered sum-of-squares decomposition assigns row-mean and column-mean deviations to network and decoder, with residual $E_{ij} = Y_{ij} - \bar{Y}_i - \bar{Y}_j + \bar{Y}$; components sum to total centered SS. MIDD A and physical-grating $Q = (\bar{u}_{\text{right}} - \bar{u}_{\text{left}})/2$ are analyzed separately; early/late windows and $|Q|$ are sensitivity endpoints. Ten thousand joint-identity bootstrap draws preserve matched cells. Intervals are conditional robustness summaries, not causal variance components; with one observation per pair, residual and measurement error cannot be separated. Within-column row and within-row column permutations are alignment diagnostics. Gratings use three spatial cycles, one temporal cycle/s, contrast 0.35, 50 frames at $\text{dt}=0.02$ s after 1-s gray preparation. Correct directions require mean rightward $u > 0$ and leftward $u < 0$ over the original grating receptors at frames 10–49. Both-direction accuracy differs from $Q > 0$; original Q is checked

first. Nine mismatches are summarized within network before matched/mismatched comparisons.

Spatial and matched-static controls

Matched-static controls use all 50 official models and ten histogram-preserving sector shuffles per family (seed 2026100301). Luminance is permuted within equal raster-area classes, preserving sector and image histograms. Canonical and shuffled magnitudes use the same image/full-LR half-difference; shuffles are averaged within model, with ten-draw resolution for ranks/percentiles. Mean/median differences and 10,000 whole-model percentile intervals use seed 2026100504. Other controls are histogram-matched random pixels, uniform gray 128, and 382×382 static/CW/CCW propellers. Operators remain distinct: Original/Mirror, image/LR, CW/CCW, and singleton uniform T/G.

The primary phase analysis tests $C=48$ in all 50 official models at twelve equally spaced phases $j/12$ of the 15° color-unit period. $C=207/4$ MIDD phase tests, placement, and radial profiles use official models 000–009. These phases span half the balanced negative’s 30° period. $ICC(A,1)$ is the absolute-agreement single-measure intraclass correlation; it estimates the share of model-by-phase variance attributable to between-model differences, with phase and residual variance contributing to the remainder. $ICC(A,1)$ and consistency ICC use the $\text{model} \times \text{phase}$ matrix; 10,000 whole-model bootstrap intervals are conditional on this grid. Phase-wise reversal and reversal of phase-averaged raw pairs are distinct endpoints. Fixed ± 3 -pixel translations use white fill, no interpolation/wrapping, and a fixed retinal annulus. Pixel hashes identify exchanged/duplicate reflection/rotation stimuli; their sign changes are not independent network evidence.

Radial geometry uses actual BoxEye receptor positions. Euclidean nearest-neighbor distances determine local spacing; unit-period arc is $r \times 2\pi/24$. Seven equal radial bins span 0.14–0.98 r_{max} . T_O, T_M, A, S, G and reversal are recomputed from saved flow within each bin and the outer subset satisfying period $\text{arc} \geq 2 \times \text{local spacing}$. Receptors are equally weighted within each ROI and analyzed frames 10–49. Full-annulus recalculation is checked against source scalars before ROI comparisons. The geometric criterion does not ensure sharp-edge sampling or isolate aliasing; these comparisons do not establish resolution invariance.

Sector counts were 36,48,60,72,84,96,120 for 3MIDD and 32,48,64,72,80,96,120 for 4MIDD, at phase fractions 0 and 0.5 of one color unit. Counts maintain integer complete units and balanced negatives. The count cohort was the fixed pilot 008/009/005 followed by 000–004/006/007 ($N=10$). Image dimensions, annulus, and luminances were unchanged; edge density covaries with sector count. Within-model normalized response contrasts and Spearman summaries are bootstrapped across models, never across sectors.

Held-out scalar prediction and physical calibration

Conditional leave-one-luminance-out fits exclude all A4 pairs containing the held luminance; independently measured A3 values remain available to form predictions. Each of 21 pairs is predicted in two folds, giving 42 repeated predictions per model. Ordinary least-squares slope/intercept are fitted on training pairs only. R^2 , RMSE, and sign accuracy are summarized within model and then equally across models. The sign mask is fixed to the canonical predictor’s 5% A3-range rule; affine CV sign accuracy differs from unfitted component sign agreement. Leave-one-pair-out is secondary.

Stronger comparison predictors are own-model A3 difference, the mean A3 curve of all nine other models (target excluded), each of the nine individual donors, and C1–C2. All use identical training folds, affine fits and the original target-model sign mask. Donor metrics are averaged within target

before inference; 90 donor pairs do not provide 90 independent units. Primary mean CV- R^2 advantages use 10,000 model-bootstrap draws and exact 2^{10} paired sign-flip tests, with Holm adjustment across three contrasts (seed 2026100502). Secondary metrics include RMSE, A3-range-normalized RMSE and sign accuracy.

Additional comparison predictors comprise the A3 sum, assignment permutations, and random antisymmetric potentials B(C1)–B(C2). On this complete grid, A3-value and luminance-label permutations are the same null. Smooth Gaussian-process potentials use squared-exponential length-scale 0.25 as primary and 0.1/0.5 as sensitivities, centered and RMS-matched within model. We use 2,000 shared null draws and 10,000 model-bootstrap draws, seed 2026100305, with Holm adjustment of four ensemble null-rank comparisons. Rank p is $(1 + \text{number of null draws at least as extreme}) / (2000 + 1)$; the minimum is $1/2001$ and the four-comparison Holm floor is $4/2001$.

Physical CW/CCW/static propeller movies were evaluated in the exact centered eight-field geometry in official 000–009, alongside a separate native rendering. Rendered geometry and global coordinate merging followed the stated method. Calibration confirms the implemented scalar sign without altering stored data; native and multi-field amplitudes are not treated as matched perceptual effect sizes.

Matched decoder-input replacement and prolonged stimulation

The common replacement experiment used official 000–009, reusing the three pilot models exactly. The gray-0.5 baseline was frame-, sample-, cell-channel-, and receptor-matched. Only targeted decoder-input channels were replaced, with no recurrent re-simulation. Groups were T4a, T4b, T4, T4+T5, Tm1 sham, and all decoder inputs. No-op equality and nontarget invariance were verified. Seven 3MIDD luminances 0,48,85,128,170,207,255 in Original/Mirror pairs, propeller static/CW/CCW, and right/left moving edges were evaluated with the same operation. Its gray baseline and luminance support differ from the 21-condition mean-activity replacement, so estimates are not pooled.

The primary metric is signed intervention-minus-noop contrast divided by the paired noop mean flow magnitude G . MIDD contrast is the $C=48$ Original/Mirror half-difference; physical contrasts are CW/CCW and right/left half-differences. Neither noop A nor intervention magnitude is the denominator. Only nonfinite/nonpositive G or invalid provenance is excluded; no such model was excluded. Secondary attenuation multiplies the normalized change by minus the noop contrast sign. Raw signed changes, medians, leave-one-model-out means, seven-luminance projected gain $(y \cdot x) / (x \cdot x)$, and RMS-normalized profile differences remain separate endpoints. Broad T4 and T4+T5 are primary; T4a is secondary. Whole-model percentile intervals use 10,000 draws (seed 2026100502); these stimulus comparisons do not claim dedicated selectivity. The separate 21-condition-baseline profile analysis retains its own statistics.

For prolonged presentation, official 024/041/042 were fixed from 90th/50th/10th canonical $C=48-|A|$ quantile representatives. Each run simulated 500 consecutive frames at 50 Hz without resetting between windows. Static fade-in durations were 0,0.5,1,2 s; $\pm 1/\pm 2$ -pixel translation jitter used the 1-s fade only. One balanced path with seed 2026100414 was shared across conditions/models: first and last positions were zero and the remaining 498 frames balanced. Negative and uniform controls were included. Predefined windows were frames 10–49,50–99,200–249,450–499. Late/early and fade/jitter contrasts are repeated measurements in three selected models; the 0.0005 floor is operational, not a biological threshold.

Fixed-checkpoint static controls use all fourteen custom seeds at steps 0,10k,50k,100k, selected in-

independently of response. C=48 Original/Mirror, balanced negative/LR, one fixed sector shuffle/LR and uniform128 are measured with the original custom runtime. Archive-member identities and stored response values are checked. Uniform is an unpaired tangent output, never an invented anti-symmetric response. Exact two-sided binomial tests at step 100,000 compare positive/negative seed counts with probability 0.5; the three condition-level tests are reported with Bonferroni adjustment, with raw values given in Results. Adjacent-checkpoint and odd/even late 80–100k summaries are descriptive repeated-measure checks, not independent seeds or evidence of convergence.

Sparse physical-motion training controls used custom seeds 000/004/007 at steps 0–100,000 every 10,000 steps, selected before inference. Propeller direction/detection criteria and moving-edge signed contrasts were compared with dense 1,000-step MIDD records. Persistent crossing requires consecutive available evaluation checkpoints; intervals reflect the different sampling resolutions. No precise universal onset ordering is inferred. Learning threshold uncertainty uses 10,000 seed-bootstrap draws with seed 2026100308, and the sparse physical-control protocol uses seed 2026100309.

Statistical analysis and reproducibility

The analysis unit is model or learning seed; conditions, pairs, positions, and checkpoints are repeated/structured observations. Figures show model means $\pm$ SEM or seed medians/IQR as indicated; thin curves show individual trajectories. Sign counts and supplementary ensemble direction agreement are descriptive; the latter uses the same-condition 50-model mean A as reference and is not an independent test of human direction.

Supplementary Fig. S9 reports Benjamini–Hochberg-adjusted activity–flow coupling and corrected-family decoder-input replacement mean losses, 95% bootstrap intervals, sign-flip p values, and family-adjusted q values. T4b remains exploratory. Recurrent whole-subtype EPE and subtype changes are descriptive and are not pooled with decoder-input replacement estimates.

Figures use stored responses, summaries, and the Fig. 1B raster snapshot; assembly requires no new inference. Model sets, rendering, time windows, and statistics are specified for each analysis.

Data and code availability

Code, figure notebooks, processed inputs, and reproducibility metadata are available in the flyvis-midd repository. Saved analysis artifacts support figure rendering without rerunning acquisition notebooks or model inference (Table S4). FlyVis v1.2.0, commit 92b3845cc426dd309a1a0e1b3890156c42e14021, and official pretrained models are distributed separately through FlyVis (flyvis download-pretrained). Custom learning checkpoints are provided at data/checkpoints/learning_custom_checkpoints.zip.

The PDF includes Supplementary Tables S1–S5B; S5A/B summarize T4c phenotype, T4/T5 activity, and ON/OFF source currents. Model-level records and provenance are under data/precomputed/supplementary/biological_constraints/. Fig. 1B uses frozen raster snapshots because complete flow arrays for all displayed models were not stored.

Supplementary text

Additional biological and circuit constraints

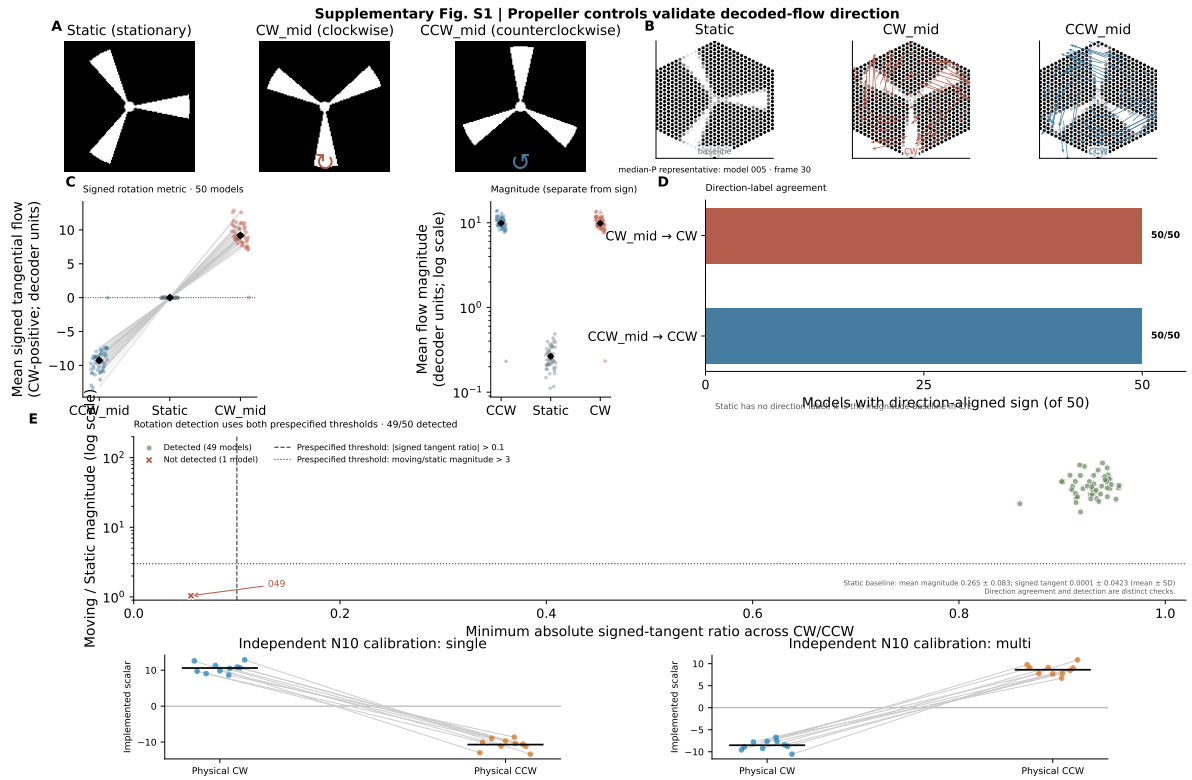
Saved T4c phenotype labels were compared with phase-averaged 3MIDD C=48 responses in the official 50-model set. Mean signed A was -0.007707 for upward-selective ($n=31$), $+0.002674$ for downward-selective ($n=7$), and $+0.002613$ for non-direction-tuned ($n=12$) models. The cluster comparison was suggestive but inconclusive after correction (eta-squared=0.157, permutation $p=.0176$, Holm $p=.0528$); phase-instability and sign-change comparisons were also inconclusive (Holm $p=.684$ and $.301$; Table S5A). No independent broader biological-response score was available, so no composite fidelity measure was constructed. This association does not establish T4c causation or general prediction of MIDD output.

In the official 50-model cohort, the fixed equal-weight T4/T5 activity projection did not establish a common directional readout or model-level association with decoded A : the mean activity component was $+0.000141$ (95% CI $[-0.000677, +0.001055]$, Holm $p=1.000$), its association with decoded A was Spearman $\rho=.116$ (95% CI $[-.193, .415]$, Holm $p=1.000$), and sign agreement was 27/50 (54%; Table S5B). This is inconclusive for the tested fixed aggregate readout and is not evidence that T4/T5 activity contains no MIDD-related signal.

A separate broad ON-minus-OFF source-current summary was likewise inconclusive: mean I_{ONOFF} was 4.52×10^{-8} (95% CI $[-1.12 \times 10^{-7}, 2.16 \times 10^{-7}]$, Holm $p=1.000$), its association with decoded A was Spearman $\rho=.065$ (95% CI $[-.216, .341]$, Holm $p=1.000$), and descriptive sign agreement was 26/50 (52%; Table S5B). A secondary OFF_A association with decoded A was negative (Spearman $\rho=-.357$, Holm-adjusted $p=.048$ within the secondary family), but remains a secondary observational result and does not identify a pathway mechanism. No D2 causal intervention was performed.

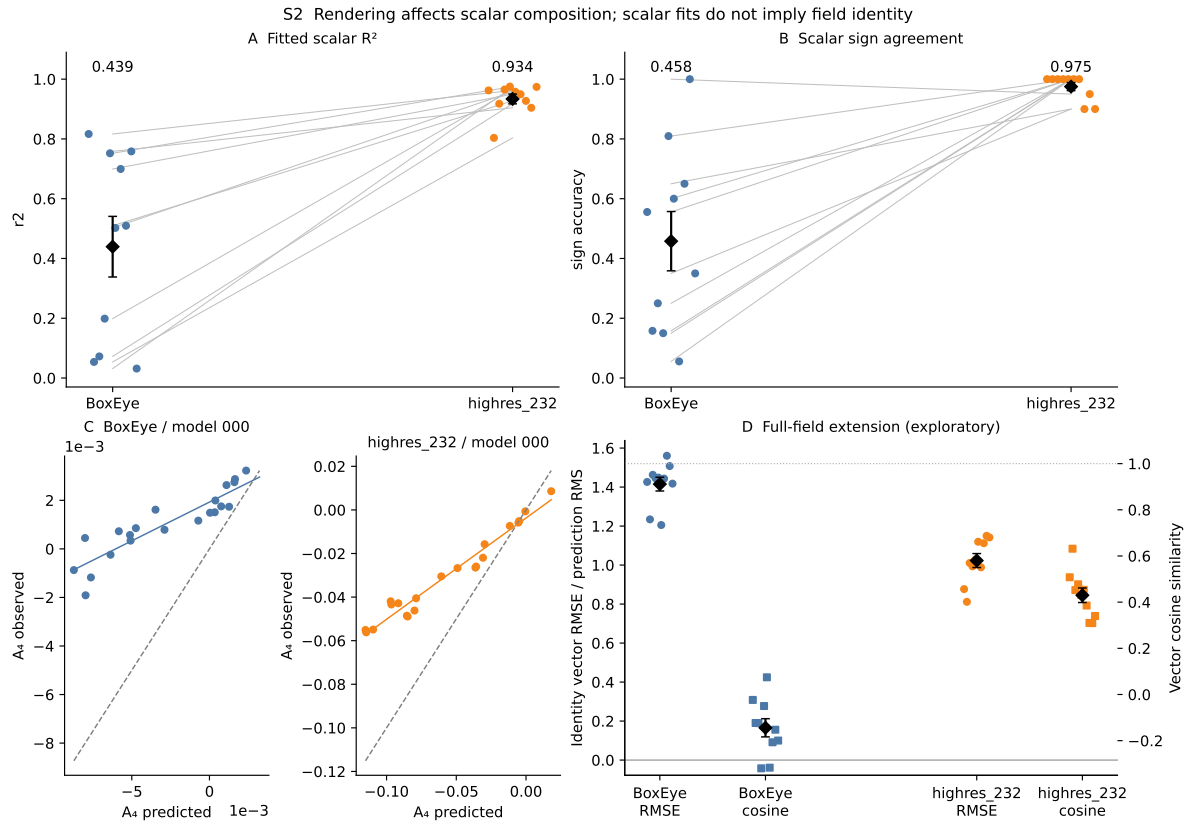
Supplementary figure legends

Supplementary Figure S1. Physical-motion sign calibration



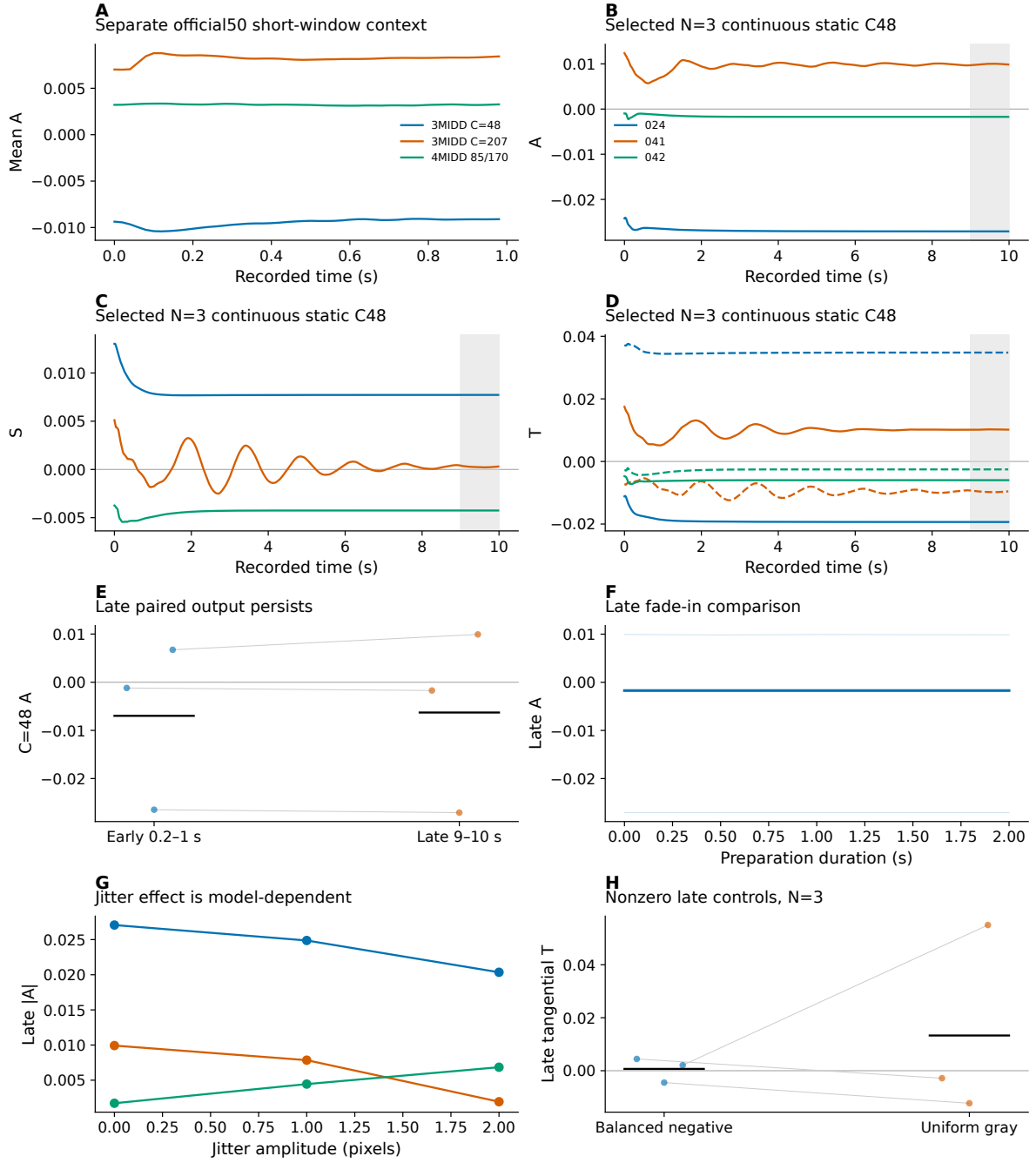
Native propeller stimuli: official models 000–049 (N=50), 128×128 pixels, 50 frames at 50 Hz, analyzed over frames 10–49. Expected direction was obtained in 50/50 models; the additional detection criteria were met in 49/50. Centered eight-field calibration: models 000–009 (N=10); CW scalars were negative and CCW scalars positive in all ten models. The native basis has the opposite sign.

Supplementary Figure S2. Rendering comparison



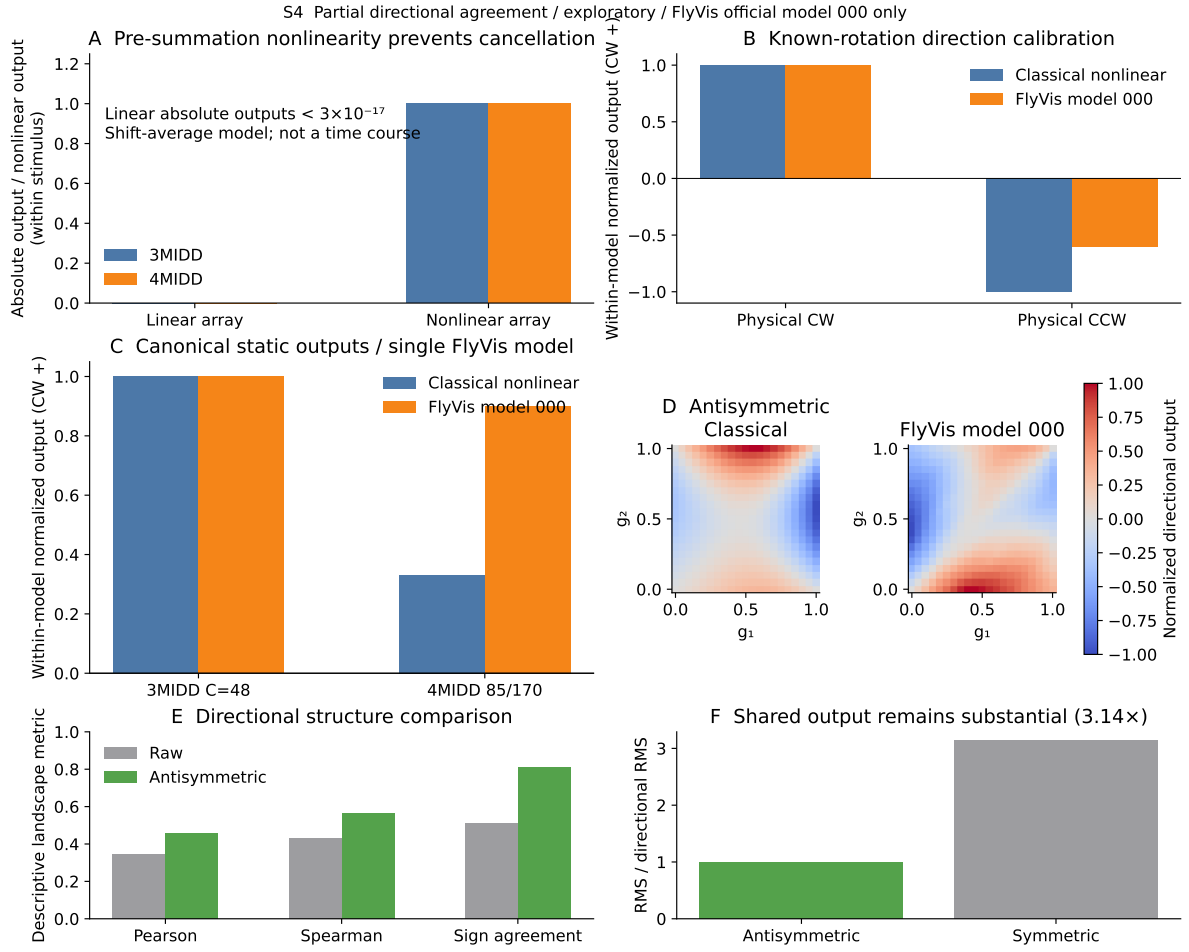
Official models 000–009 ($N=10$), BoxEye and highres_232 seven-field renderings. **A**, model-wise fitted scalar R^2 for observed A_4 versus $A_3(C1)–A_3(C2)$. **B**, sign agreement. Gray lines pair models; black diamonds and bars show means $\pm$ SEM. **C**, predicted and observed scalar responses for model 000. **D**, full-field vector RMSE and cosine similarity, summarized within model across 21 luminance pairs.

Supplementary Figure S3. Prolonged static presentation



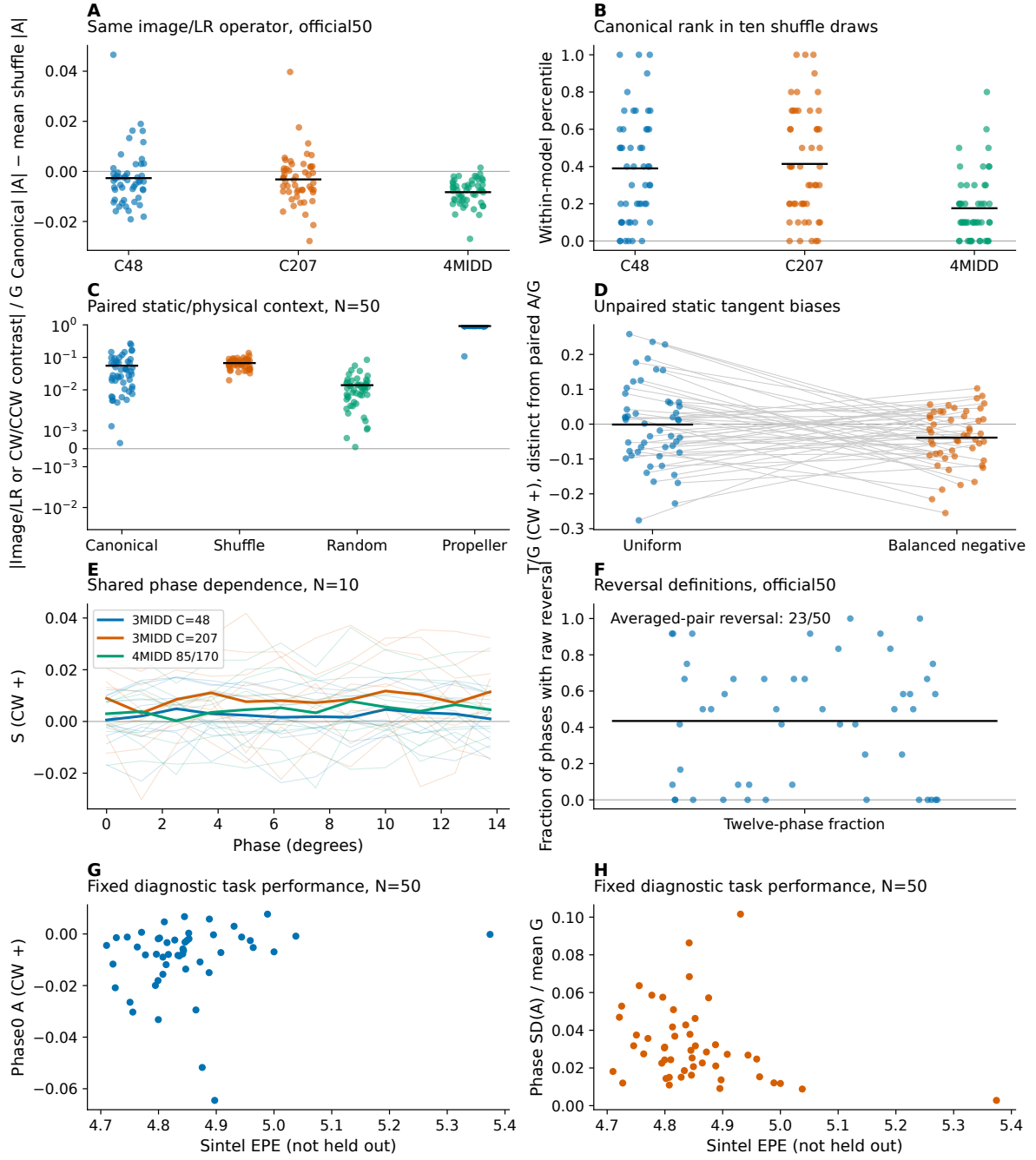
Three selected official models (024, 041, 042), 500 frames at 50 Hz. Panels show C=48 A, S, and paired raw T over time, early and late windows, and repeated fade-in and translation-jitter conditions. Individual-model traces are shown; late A values are -0.02707 , $+0.009925$, and -0.001720 .

Supplementary Figure S4. Classical nonlinear motion-detector comparison



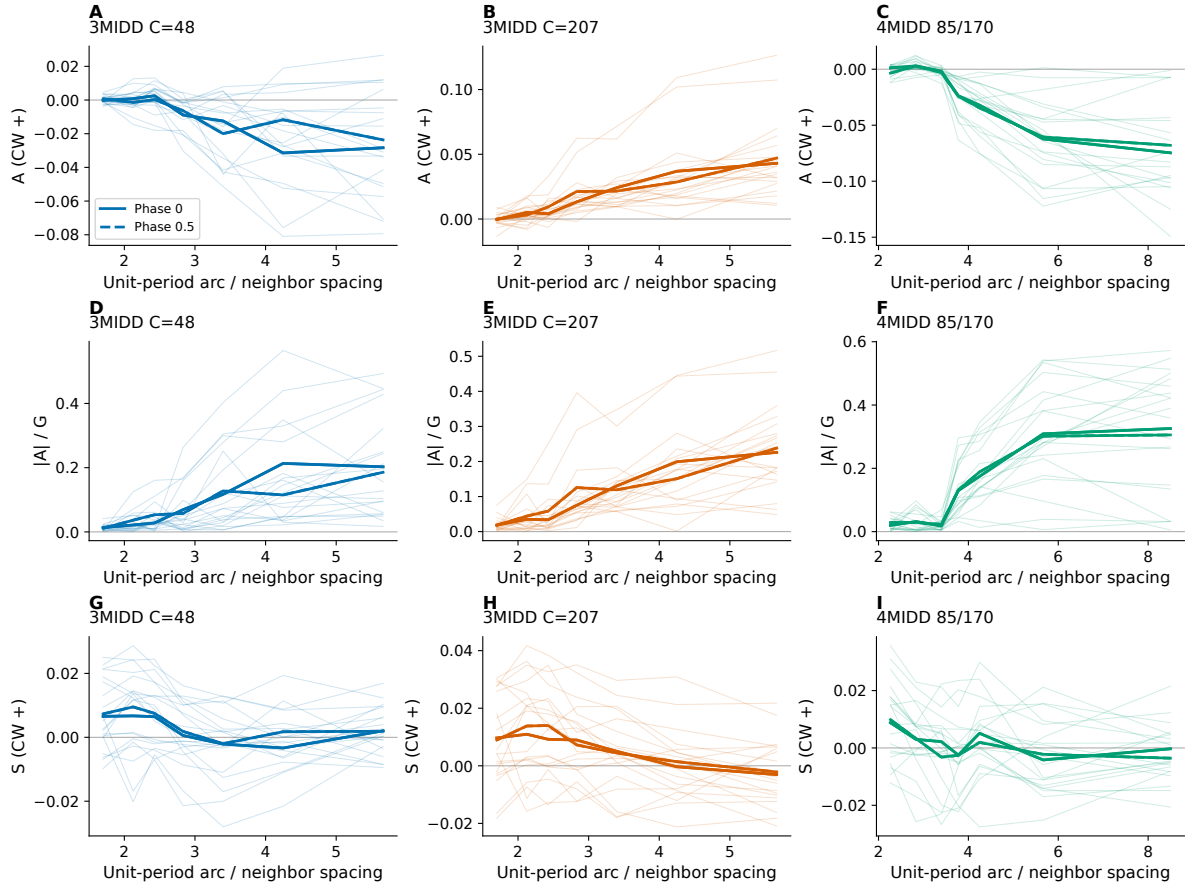
Exploratory comparison of a Reichardt-array model and FlyVis model 000. The array applies $f(d) = \sqrt{2d}/\sqrt{1+d^2}$ before spatial summation (Bach and Atala-Gérard 2020). **A–C**, normalized outputs for linear/nonlinear detector variants, physical CW/CCW calibration, and static C=48/4MIDD 85–170 stimuli. **D–F**, normalized four-level response landscapes and directional/shared components. FlyVis uses native BoxEye and frames 10–49; no ensemble statistics are shown.

Supplementary Figure S5. Static controls and phase context



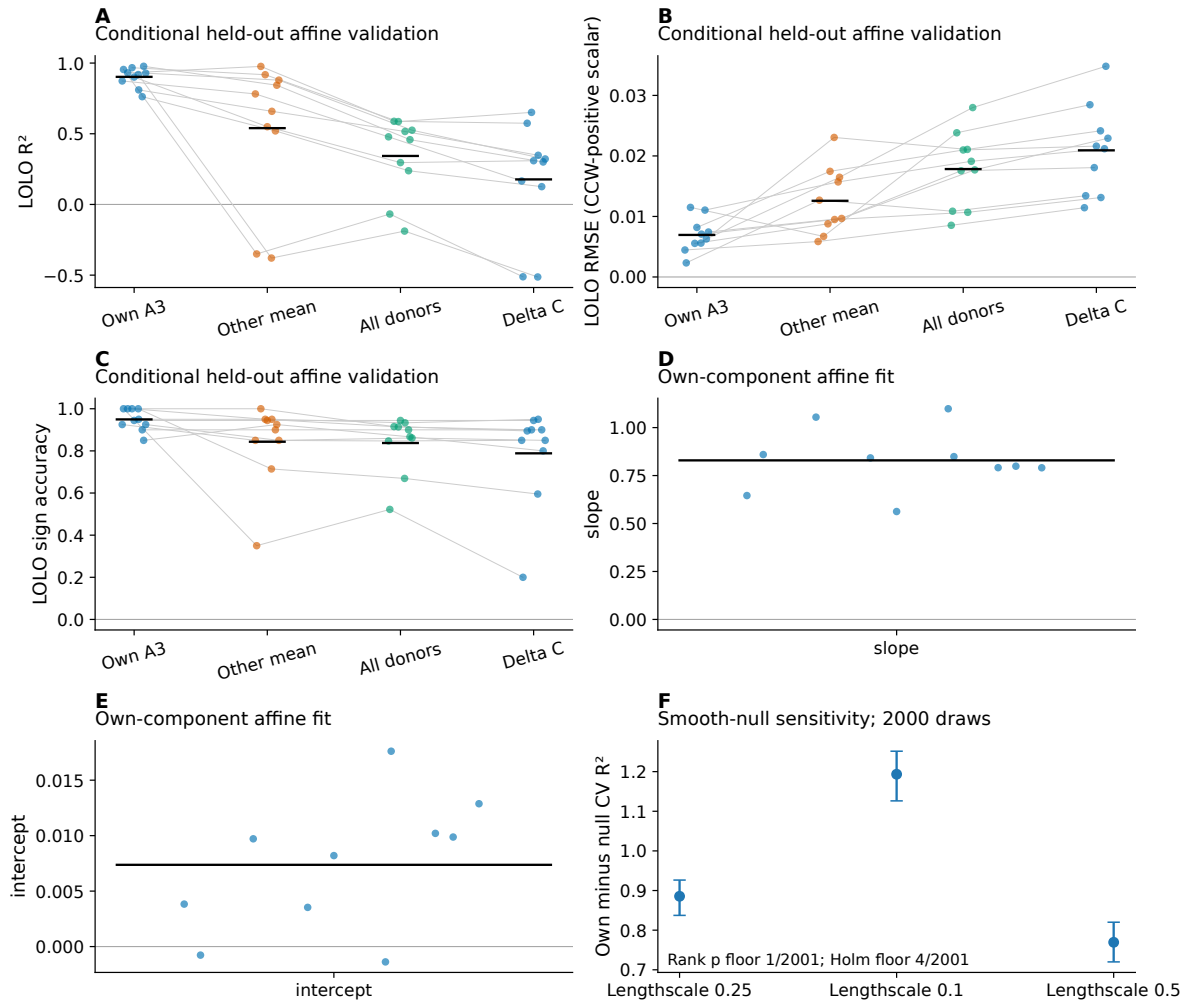
Official models 000–049 (N=50), native rendering. **A–D**, sector shuffles, canonical/random image-reflection pairs, propellers, uniform and negative controls. **E**, S across phase in the separate ten-model analysis. **F**, phase-wise and phase-averaged reversal summaries. **G–H**, diagnostic Sintel EPE against phase-zero A and phase variability.

Supplementary Figure S6. Sector-count dependence



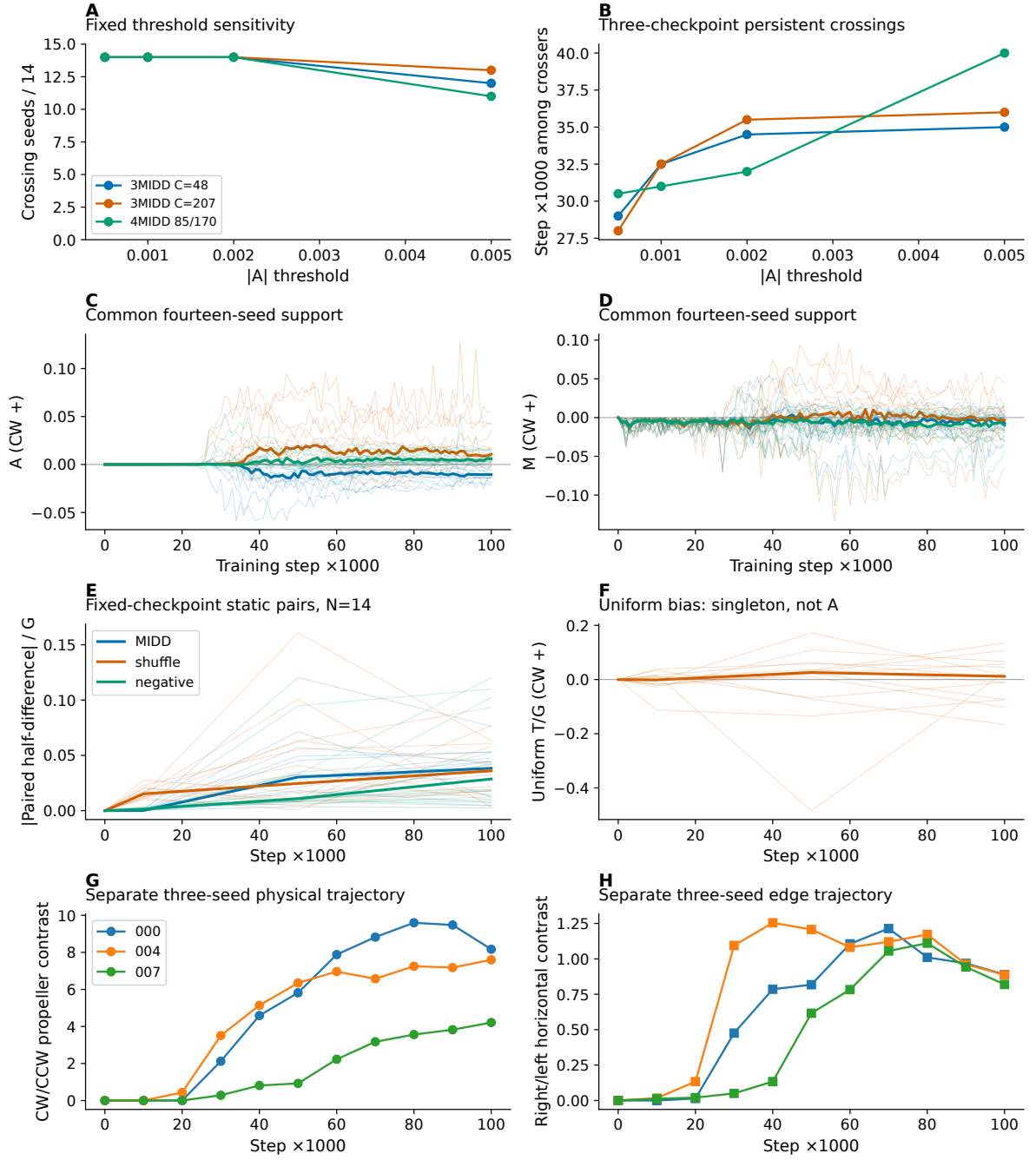
Separate official ten-model analysis; 3MIDD 36–120 and 4MIDD 32–120 sectors at phase fractions 0 and 0.5. Columns show $C=48$, $C=207$, and 4MIDD; rows show A , $|A|/G$, and S . Thin lines denote models and heavy lines show model means. The horizontal axis scales unit-period arc by receptor spacing.

Supplementary Figure S7. Scalar validation and comparison predictors



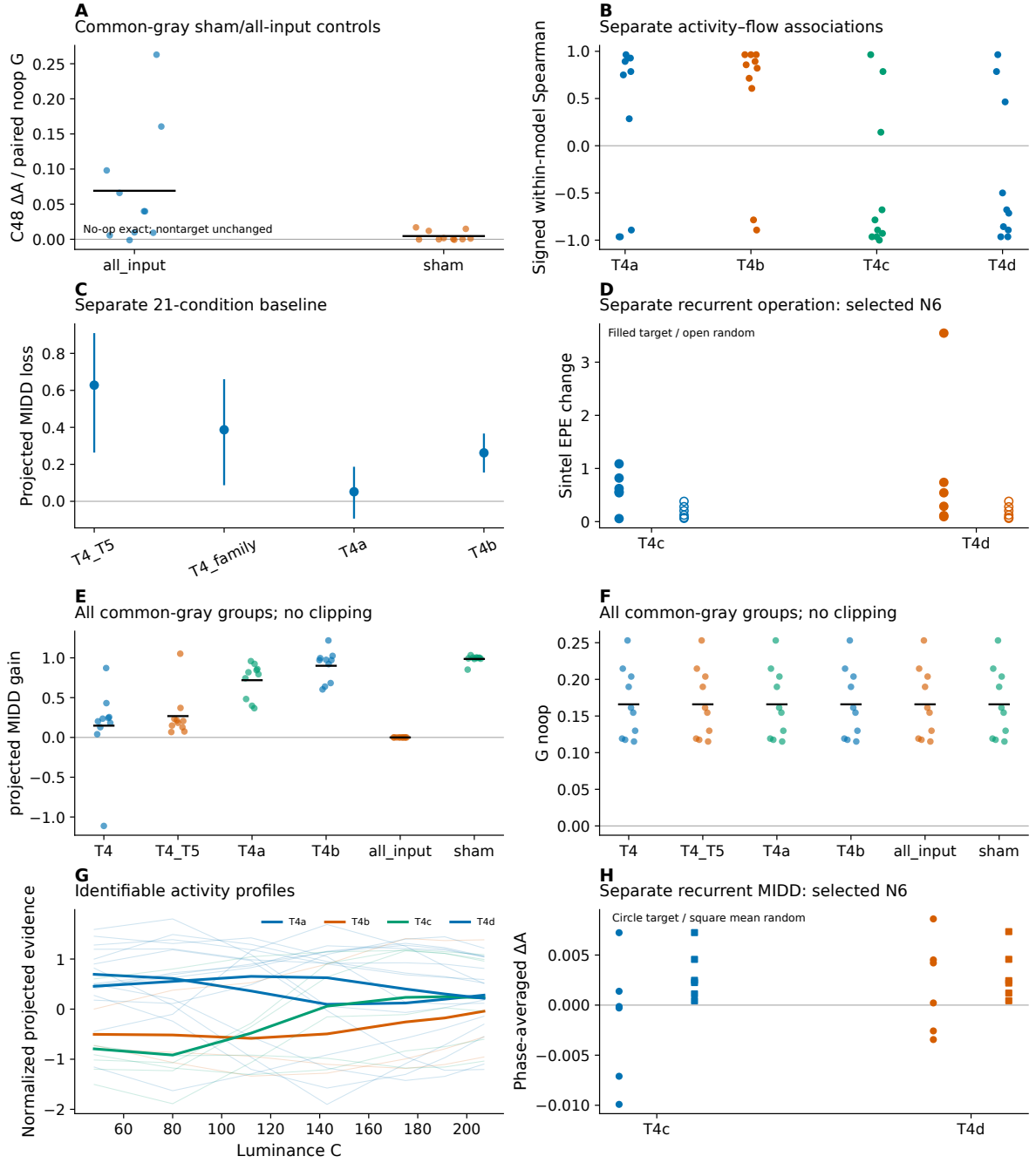
Official models 000–009, centered eight-field rendering, CCW-positive scalar. **A–C**, conditional leave-one-luminance-out R^2 , RMSE, and sign accuracy for the listed predictors. **D–E**, model-wise slopes and intercepts. **F**, own-model versus smooth-null $CV-R^2$. Intervals are conditional model-bootstrap intervals.

Supplementary Figure S8. Learning response and physical-motion controls



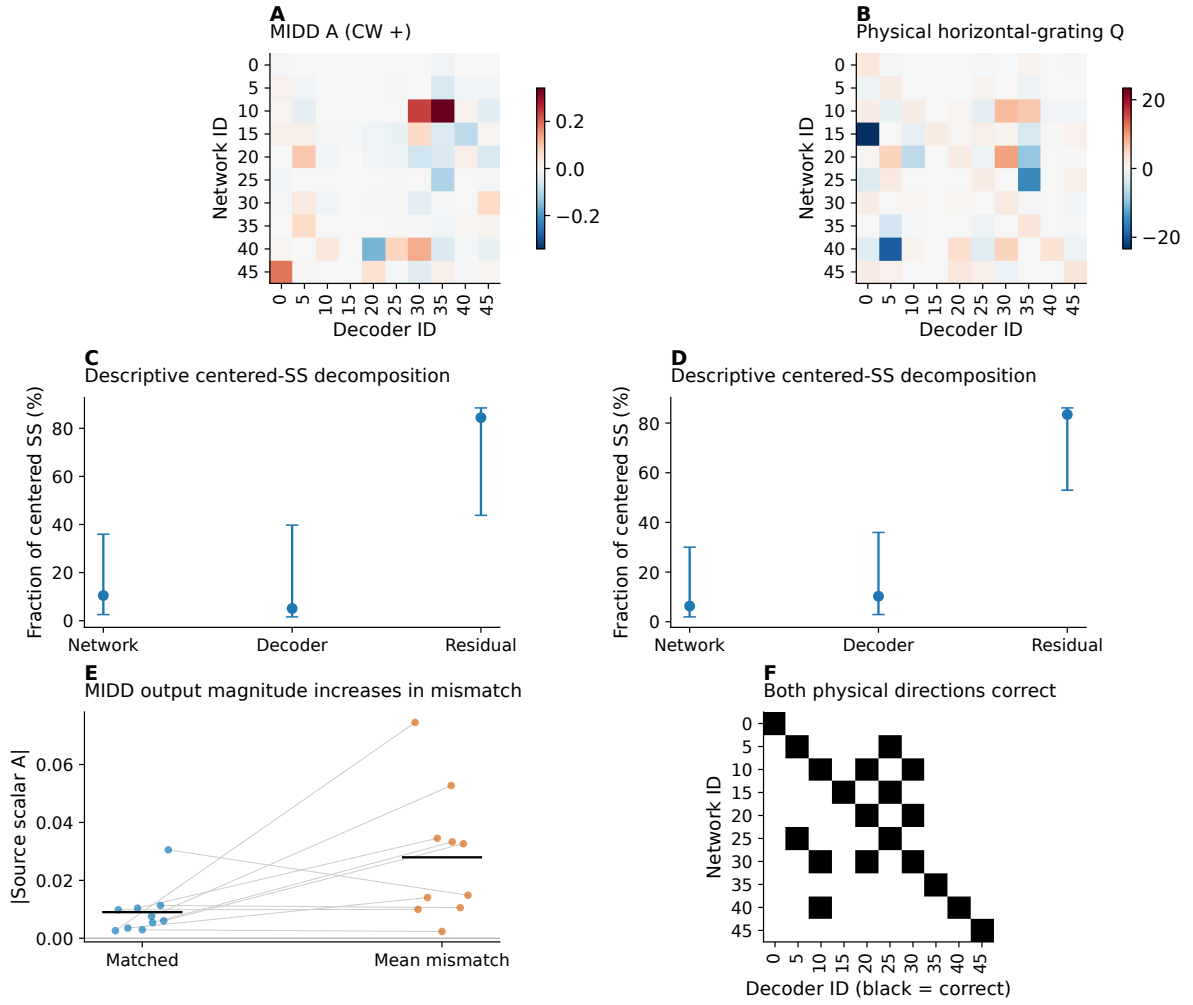
Learning-custom seeds 000–013 at the common 100,000-step endpoint. **A–B**, first and persistent threshold crossings. **C–D**, signed A and reversal margin trajectories. **E–F**, MIDD, shuffle, negative, and uniform responses at steps 0, 10,000, 50,000, and 100,000. **G–H**, propeller and moving-edge trajectories for seeds 000, 004, and 007.

Supplementary Figure S9. Intervention and activity controls



Panels show sham and all-input replacements, activity-flow associations, the separate 21-condition decoder-input analysis, recurrent subtype perturbations and controls, denominator checks, and subtype activity curves. Each analysis retains its own stimulus and intervention definition; intervals and corrected statistics are reported in the source tables.

Supplementary Figure S10. Network–decoder cross-swap



Selected official models crossed in a 10×10 design. **A**, $C=48$ MIDD A matrix. **B**, physical horizontal-grating response Q . **C–D**, network, decoder, and residual fractions with 10,000 joint-identity bootstrap intervals. **E**, matched and mismatched $|A|$. **F**, rightward and leftward direction accuracy.

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Supplementary numerical tables

Table S1. Cohorts and measurement conditions

analysis	cohort_units	renderer_sign	support	boundary
Canonical baseline	official 000–049; 50	native; CW-positive	50 frames; 10–49	stimulus-first 1 s fade
Matched shuffled controls	015/025/046+000–006; 10	native; CW-positive	50 frames; 10–49	restricted exact-raster histogram shuffles; 10 repeats
Phase/reflection	000–009; 10	native; CW-positive	12 phases; 8 transforms	duplicate images disclosed; fixed retinal ROI
Sector count	008/009/005+000–004/006/007; 10	native; CW-positive	7 counts; 2 phases	spatial frequency covaries with edge density
Scalar composition/calibration	000–009; 10	centered 8-field; positive CCW	21 pairs; LOLO 42 repeated predictions	independent held-luminance A3 available
Rendering comparison	000–009; 10	native versus seven-field	source-specific scalar/full-field	distinct from centered 8-field
Learning	custom 000–013; 14	native; CW-positive	common 0–100k	unequal follow-up; EPE diagnostic
Physical learning controls	custom 000/004/007; 3	native	11 checkpoints at 10k spacing	sparse physical versus dense MIDD
Prolonged static	official 024/041/042; 3	native; CW-positive	continuous 500 frames; late 450–499	phenotype-selected; one jitter path
Common replacement	official 000–009; 10	native; gray 0.5 baseline	same static/physical operation	not historical 21-condition replacement
Recurrent intervention	six selected custom instances	recurrent state neutralization	paired Sintel/MIDD; 199 random draws	002 holdout designation; not prospective confirmatory
Cross-swap	official 000: 005: 045; 10×10	A vs physical horizontal Q	historical gray; stable 10–49	conditional descriptive SS; one observation per pair
Phase and static-control analysis	official 000–049; 50	native; CW-positive	12 C48 phases; ten shuffles/family	canonical fade 1 s; frames 10–49; operators explicitly separated
Scalar-composition comparator analysis	official 000–009; 10	centered 8; CCW-positive	7 luminances; 21 pairs; 42 fold predictions	target excluded; all 9 donors averaged within target
Radial receptor-sampling analysis	official 000–009; 10	native; CW-positive	7 bins; 12 phases; three families	actual spacing 12–13 px; full 698/outer 532
Stable decoder-input intervention	official 000–009; 10	native endpoints; class-specific contrasts	same common-gray replacement	signed change / paired noop G; T4/T4+T5 primary

analysis	cohort_units	renderer_sign	support	boundary
Official-cohort task-performance diagnostic	official 000–049; 50	native flow vectors; EPE	15 fixed training/final samples; 40 resampled frames	same custom Fig 5 pipeline; not held out
Fixed-checkpoint static controls	custom 000–013; 14	native; CW-positive	steps 0/10 k/50 k/100 k	one fixed shuffle; uniform singleton; paired operators separated
Cross-swap physical-direction validation	ten joint identities; 100 cells	rightward u positive; left negative	stable 10–49 grating; 1 s gray	both individual signs required; cells not independent n

Table S2. Summary of model-level analyses

Primary estimates are reported in the main text and figure legends. Complete model- and seed-level estimates, intervals, and secondary endpoints remain available in `figures/supplementary/tables/TableS2_model_unit_estimates.csv` and the companion individual-estimate CSVs. Source records include `official50_performance_primary_relations` and `comparator_summary`; repeated conditions remain grouped within models.

analysis source	records
<code>matched_control_bootstrap_summary.csv</code>	15
<code>official50_shuffle_summary.csv</code>	15
<code>comparator_contrasts.csv</code>	3
<code>comparator_summary.csv</code>	16
<code>composition_bootstrap_summary.csv</code>	8
<code>geometry_bootstrap.csv</code>	120
<code>official50_phase_reliability.csv</code>	1
<code>phase_bootstrap.csv</code>	24
<code>learning_bootstrap_summary.csv</code>	15
<code>learning_stability_sign_tendency_100k.csv</code>	3
<code>official50_performance_primary_relations.csv</code>	3
<code>denominator_audit.csv</code>	18
<code>replacement_bootstrap.csv</code>	32
<code>robust_intervention_summary.csv</code>	90
<code>duration_bootstrap.csv</code>	90
<code>sector_bootstrap.csv</code>	30

Table S3. Crossed network/decoder summary

Fractions are descriptive; residual combines interaction-like departures and inseparable error. Complete estimates and direction records remain in the companion CSV files.

group/window	metric	effect	fraction	95% interval
3MIDD_048 / early	A	network	0.0614256540600169	to
3MIDD_048 / early	A	decoder	0.0688027105505688	to
3MIDD_048 / early	A	residual	0.8697716353894143	to
3MIDD_048 / early	S	network	0.0404678336667362	to
3MIDD_048 / early	S	decoder	0.1590623052445388	to
3MIDD_048 / early	S	residual	0.8004698610887248	to
3MIDD_048 / late	A	network	0.1219293662737013	to
3MIDD_048 / late	A	decoder	0.0521282869262262	to
3MIDD_048 / late	A	residual	0.8259423468000725	to
3MIDD_048 / late	S	network	0.0164764772473201	to
3MIDD_048 / late	S	decoder	0.1550631177886699	to
3MIDD_048 / late	S	residual	0.8284604049640101	to
3MIDD_048 / stable	A	network	0.1045415847416332	0.0253597628228532 to 0.3597190671914368
3MIDD_048 / stable	A	decoder	0.0508303905102007	0.0156295086357082 to 0.3975587874454097
3MIDD_048 / stable	A	residual	0.8446280247481659	0.4380382461313962 to 0.8848715310698949
3MIDD_048 / stable	S	network	0.0136036324218209	0.0044932589863485 to 0.3463091839581664
3MIDD_048 / stable	S	decoder	0.1425637376377033	0.0240842323539483 to 0.4109316602372039
3MIDD_048 / stable	S	residual	0.8438326299404756	0.4803980150483881 to 0.9014504370126556
3MIDD_207 / early	A	network	0.2078662322210796	to
3MIDD_207 / early	A	decoder	0.1501326302407385	to
3MIDD_207 / early	A	residual	0.6420011375381818	to
3MIDD_207 / early	S	network	0.0220993125871304	to
3MIDD_207 / early	S	decoder	0.1472712605083927	to
3MIDD_207 / early	S	residual	0.8306294269044763	to

group/window	metric	effect	fraction	95% interval
3MIDD_207 / late	A	network	0.2101679099689782	to
3MIDD_207 / late	A	decoder	0.1155584345388352	to
3MIDD_207 / late	A	residual	0.6742736554921868	to
3MIDD_207 / late	S	network	0.0278918725676226	to
3MIDD_207 / late	S	decoder	0.1779461746108172	to
3MIDD_207 / late	S	residual	0.7941619528215604	to
3MIDD_207 / stable	A	network	0.0606196244842266	0.0137383918774579 to 0.3469018415141806
3MIDD_207 / stable	A	decoder	0.1534466403569925	0.0471938269786996 to 0.3855226398561193
3MIDD_207 / stable	A	residual	0.7859337351587808	0.4458490398169142 to 0.8640462789035956
3MIDD_207 / stable	S	network	0.0257832898577366	0.0074338402882862 to 0.3306083392778729
3MIDD_207 / stable	S	decoder	0.165911640494224	0.0336990195207977 to 0.3986384896431253
3MIDD_207 / stable	S	residual	0.8083050696480397	0.4785150046785157 to 0.898163994655744
4MIDD_085_170 / early	A	network	0.042560141257751	to
4MIDD_085_170 / early	A	decoder	0.141161981620001	to
4MIDD_085_170 / early	A	residual	0.816277877122248	to
4MIDD_085_170 / early	S	network	0.0177410899613698	to
4MIDD_085_170 / early	S	decoder	0.1292789229623777	to
4MIDD_085_170 / early	S	residual	0.8529799870762521	to
4MIDD_085_170 / late	A	network	0.1722303625102234	to
4MIDD_085_170 / late	A	decoder	0.0965413491472198	to
4MIDD_085_170 / late	A	residual	0.7312282883425567	to
4MIDD_085_170 / late	S	network	0.021023056710978	to
4MIDD_085_170 / late	S	decoder	0.1635022035870863	to
4MIDD_085_170 / late	S	residual	0.8154747397019355	to
4MIDD_085_170 / stable	A	network	0.1435620817626706	0.0377136424225975 to 0.414913444475051

group/window	metric	effect	fraction	95% interval
4MIDD_085_170 / stable	A	decoder	0.0897671333579483	0.0235923292037321 to 0.337375356854991
4MIDD_085_170 / stable	A	residual	0.7666707848793808	0.4419084004757377 to 0.8795454816942961
4MIDD_085_170 / stable	S	network	0.0187586277249247	0.0060656639675744 to 0.3396974745942585
4MIDD_085_170 / stable	S	decoder	0.1527930489887532	0.0244251615149438 to 0.4088727028651961
4MIDD_085_170 / stable	S	residual	0.8284483232863217	0.4810528631749313 to 0.9039772176843336
gray / early	gray_tangent	network	0.0404988282658878	to
gray / early	gray_tangent	decoder	0.1552584562874993	to
gray / early	gray_tangent	residual	0.804242715446613	to
gray / late	gray_tangent	network	0.0229019926438021	to
gray / late	gray_tangent	decoder	0.16681681260484	to
gray / late	gray_tangent	residual	0.8102811947513574	to
gray / stable	gray_tangent	network	0.0248160228122199	0.0077778425861428 to 0.3379363597794899
gray / stable	gray_tangent	decoder	0.1645652337593056	0.0293927948896213 to 0.4197970007759997
gray / stable	gray_tangent	residual	0.8106187434284743	0.4793755081447476 to 0.8883983332447825
motion_grating / early	Q_motion	network	0.0566146725920646	to
motion_grating / early	Q_motion	decoder	0.1015325021889768	to
motion_grating / early	Q_motion	residual	0.8418528252189585	to
motion_grating / early	abs_Q_motion	network	0.1587144570546492	to
motion_grating / early	abs_Q_motion	decoder	0.2240053536360076	to
motion_grating / early	abs_Q_motion	residual	0.6172801893093433	to
motion_grating / late	Q_motion	network	0.0731285759342911	to
motion_grating / late	Q_motion	decoder	0.1029978493481544	to
motion_grating / late	Q_motion	residual	0.8238735747175544	to
motion_grating / late	abs_Q_motion	network	0.1097869881038825	to
motion_grating / late	abs_Q_motion	decoder	0.18389046464922	to

group/window	metric	effect	fraction	95% interval
motion_grating / late	abs_Q_motion	residual	0.7063225472468975	to
motion_grating / stable	Q_motion	network	0.0630847455258727	0.019483913769204 to 0.3002329672628942
motion_grating / stable	Q_motion	decoder	0.1026847701563924	0.0290441070615179 to 0.3594701107529393
motion_grating / stable	Q_motion	residual	0.8342304843177346	0.5296760250273576 to 0.8608622090714152
motion_grating / stable	abs_Q_motion	network	0.1233756375462799	0.0736326239777872 to 0.390347134935715
motion_grating / stable	abs_Q_motion	decoder	0.14932545635452	0.0945752647673025 to 0.3978775926903129
motion_grating / stable	abs_Q_motion	residual	0.7272989060991999	0.3719943528624801 to 0.7487237094624473

Table S4. Reproduction inventory

Complete relative paths, SHA-256 values, and reproduction routes are provided in `figures/supplementary/tables/TableS4_reproduction_inventory.csv` (115 entries).

Table S5A. T4c phenotype associations with MIDD response

Official pretrained cohort; phase observations are summarized within model. Entries are cluster means/proportions with 95% model-bootstrap intervals; omnibus p values are from the saved fixed-size label-permutation tests.

Endpoint	Cluster 0 (n=31)	Cluster 1 (n=7)	Cluster 2 (n=12)	Effect / p (raw / Holm)
Phase-mean signed A	-0.00771 [-0.013, -0.0032]	0.00267 [-0.00134, 0.00666]	0.00261 [-0.00181, 0.00713]	eta-squared=0.157; 0.0176 / 0.0528
Normalized phase SD	0.0323 [0.0268, 0.0382]	0.0368 [0.0203, 0.0563]	0.0283 [0.0166, 0.0443]	eta-squared=0.0164; 0.684 / 0.684
Any phase-wise sign change	0.613 [0.452, 0.774]	1 [1, 1]	0.667 [0.417, 0.917]	Cramér V=0.281; 0.151 / 0.301

Table S5B. T4/T5 activity and ON/OFF source-current summaries

Official pretrained cohort (N=50); phases are repeated within model. P values are adjusted within the stated primary or secondary analysis family. P3 sign agreement is descriptive/non-confirmatory; the OFF_A association is secondary.

Analysis / endpoint	Estimate/statistic	95% CI	Raw p	Adjusted p	Status
B1 P1 mean A_activity	+0.000141	[-0.000677, +0.001055]	0.755	1	Primary; inconclusive
B1 P2 association with A_flow	rho=+0.116	[-0.193, +0.415]	0.417	1	Primary; inconclusive
B1 P3 sign agreement	27/50 (54%)	[39.3%, 68.2%]	0.672	1	Primary; inconclusive
D1 P1 mean I_ONOFF	4.52e-08	[-1.12e-07, 2.16e-07]	0.606	1	Primary; inconclusive
D1 P2 association with A_flow	rho=+0.065	[-0.216, +0.341]	0.653	1	Primary; inconclusive
D1 P3 sign agreement	26/50 (52%)	[37.4%, 66.3%]	—	—	Primary; non- confirmatory
D1 secondary OFF_A association with A_flow	rho=-0.357	[-0.606, -0.051]	0.0119	0.0476	Secondary; candidate only

Table S6. Key numerical results and source files

Learning-seed p values are two-sided exact binomial tests at 100,000 steps; Bonferroni correction is across the three listed conditions. Source CSV paths are relative to `data/precomputed/`:

- Phase reliability: `fig04/official50_phase_reliability.csv`
- LOLO composition: `fig03/composition_bootstrap_summary.csv`
- Cross-swap directions: `fig06/crossswap_direction_accuracy.csv`
- Learning sign counts: `fig05/learning_stability_sign_tendency_100k.csv`

Measure	Condition / cohort	Reported value	Source key
Absolute-agreement ICC(A,1)	Fig. 4B; official-50, 12 phases	0.7647 (95% CI 0.5963–0.8678)	phase reliability
Models changing sign across phase	Fig. 4A; official-50, C=48	34/50	phase reliability
Conditional LOLO mean R ²	Fig. 3D; centered eight-field	0.9024 (95% CI 0.8588–0.9399)	LOLO composition
Both physical directions correct	Fig. 6E; matched / mismatched	10/10; 9/90	cross-swap directions
100k sign tendency	Fig. 5E; C=48	12/14 negative; p=0.01294 (Bonferroni 0.03882)	learning sign counts
100k sign tendency	Fig. 5E; C=207	13/14 positive; p=0.001831 (Bonferroni 0.005493)	learning sign counts
100k sign tendency	Fig. 5E; 4MIDD	11/14 positive; p=0.05737 (Bonferroni 0.1721)	learning sign counts